

化合物的英文命名

Nomenclature of compounds

— 无机物的命名 (Inorganic compounds)

1 元素与单质的命名

“元素”和“单质”的英文意思都是“element”，有时为了区别，在强调“单质”时可用“free element”。因此，单质的英文名称与元素的英文名称是一样的。下面给出的既是元素的名称，同时又是单质的名称。

S-block Element

IA

H Hydrogen

Li Lithium

Na Sodium

K Potassium

Rb Rubidium

Cs Cesium

Fr Francium

IIA

Be Beryllium

Mg Magnesium

Ca Calcium

Sr Strontium

Ba Barium

Ra Radium

P-block Element

IIIA

B Boron
Al Aluminium
Ga Gallium
In Indium
Tl Thallium

IV A

C Carbon
Si Silicon
Ge Germanium
Sn Tin
Pb Lead

V A

N Nitrogen
P Phosphorus
As Arsenic
Sb Antimony
Bi Bismuth

VIA

O Oxygen
S Sulfur
Se Selenium
Te Tellurium
Po Polonium

VIIA

F Fluorine
Cl Chlorine
Br Bromine
I Iodine
At Astatine

0

He Helium
Ne Neon
Ar Argon
Kr Krypton
Xe Xenon
Rn Radon

Common Transition Element

Fe : iron

Mn : manganese

Cu: copper

Zn: zinc

Hg: mercury

Ag: silver

Au: gold

2 化合物的命名

化合物的命名顺序都是根据化学式从左往右读，这与中文读法顺序是相反的。表示原子个数时使用前缀 (1)mono-, (2)di -, (3)tri- , (4)tetra - , (5)penta- (6)hexa-, (7)hepta-, (8)octa-, (9)nona-, (10)deca-, 但是在不会引起歧义时，这些前缀都尽可能被省去。

Naming metal ions (cations) for metal oxides, bases and salts

1. Single valence ions

Cation's name = Element

for example:



Sodium



Aluminum



Potassium



Calcium

2.Multivalence ions

Cation's name = Element(N)

For example:

Fe^{2+}	Iron(II)	or	Ferrous
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Fe^{3+}	Iron(III)	or	Ferric
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Cr^{2+}	Chromium(II)
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Cr^{3+}	Chromium(III)
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Mn^{4+}	Manganese(IV)
------------------	---------------

Mn^{2+}	Manganese(II)
------------------	---------------

对于有变价的金属元素，除了可用前缀来表示以外，更多采用罗马数字来表示金属的氧化态，或用后缀-ous表示低价，-ic表示高价。

如

FeO:	iron(II) oxide	或 ferrous oxide
Fe ₂ O ₃ :	iron (III) oxide	或 ferric oxide
Cu ₂ O:	copper(I) oxide	或 cuprous oxide
CuO:	copper(II) oxide	或 cupric oxide

Naming nonmetal ions (anions)

1. Monatomic anions

Anion's name = Element's root -ide

For example:

Cl⁻ Chloride

Br⁻ Bromide

I⁻ Iodide

S⁼ Sulfide

O⁼ Oxide

OH⁻ Hydroxide

CN⁻ Cyanide

H⁻ Hydride

2. Polyatomic oxyanions

(1). Acid radicals for normal **salt** (正酸根 -ate)

Anion's name = Central Element's root **-ate**

for example:

ClO_3^- Chlorate

IO_3^- Iodate

PO_4^{3-} Phosphate

NO_3^- Nitrate

SO_4^{2-} Sulfate

CO_3^{2-} Carbonate

(2). Acid radicals for **meta-salts** (亚酸根 -ite)

Anion's name = Central element's root -ite

for example:

ClO_2^- Chlorite

IO_2^- Iodite

PO_3^{3-} Phosphite

NO_2^- Nitrite

SO_3^{2-} Sulfite

(3). Acid radicals for **hypo-salts** (次酸根 -ite)

Anion's name =

Hypo- Central element's root **-ite**

for example:

ClO^- **Hypochlorite**

IO^- **Hypoiodite**

PO_2^{3-} **Hypophosphite**

(4). Acid radicals for **persalts** (高酸根Per -ate)

Anion's name =

Per-central Element's root -ate

for example:



Perchlorate



Periodate



Permanganate

Naming compounds

1. Metal oxide

Metal oxide = Cation + oxide

for example:

FeO Iron(II) oxide (Ferrous oxide)

Fe_2O_3 Iron(III) oxide (Ferric oxide)

Fe_3O_4 Ferroferric oxide

Pb_3O_4 Trilead tetroxide

Na_2O_2 Sodium peroxide

2. Nonmetal oxide

Nonmetal oxide = n-Nonmetal element + n-oxide

for example:

CO Carbon monoxide

CO₂ Carbon dioxide

SO₃ Sulfur trioxide

N₂O₃ Dinitrogen trioxide

P₂O₅ Diphosphorus pentoxide

N₂O₄ Dinitrogen tetroxide

(tetra-,mono-后缀中的a,o在最后一o之前省去)

有些物质常用俗称,如NO: nitric oxide N₂O: nitrous oxide

非金属氢化物

除了水和氨气使用俗称water, ammonia以外, 其它的非金属氢化物都用系统名称, 命名规则根据化学式的写法不同而有所不同。

(1) 对于卤族和氧族氢化物, H在化学式中写在前面, 因此将其看成与另一元素的二元化合物。

举例: HF hydrogen fluoride HCl hydrogen chloride
HBr hydrogen bromide HI hydrogen iodide
H₂S hydrogen sulfide H₂Se hydrogen selenide
H₂Te hydrogen telluride

(2) 对于其它族的非金属氢化物, H在化学式中写在后面, 可加后缀-**ane**, 氮族还可加-**ine**

举例: PH₃: phosphine或phosphane AsH₃: arsine或arsane
SbH₃: stibine或stibane BiH₃: bismuthane
CH₄: methane SiH₄: silane
B₂H₆: diborane

无氧酸

命名规则：hydro-词根-ic acid

举例：HCl: hydrochloric acid

H₂S : hydrosulfuric acid

3. Bases

Base = Metal cation + hydroxide

for example:



Aluminum hydroxide



Sodium hydroxide



Calcium hydroxide



Barium hydroxide



Cobalt(II) hydroxide

4. 盐(Salts)

(1). **正盐**(Normal salt) : 根据化学式从左往右分别读出阳离子和阴离子的名称。

Normal salt = Cation + anion

for example:



Mercury(II) sulfate



Mercury(I) sulfate



Potassium nitrate



Sodium carbonate



Sodium hypochlorite



iron(II) sulfate



potassium permanganate

(2) **酸式盐**: (Acidic salts)同正盐的读法，酸根中的H读做hydrogen，氢原子的个数用前缀表示。

Acidic salt = Cation + **hydrogen** + anion

for example:

NaHSO_4 Sodium **hydrogen** sulfate

Na_2HPO_4 Disodium **hydrogen** phosphate

NaH_2PO_4 Sodium di**hydrogen** phosphate

$\text{Ca}(\text{HSO}_4)_2$ Calcium **bisulfate**

NaHCO_3 Sodium **hydrogencarbonate**

或 Sodium **bicarbonate**

(3).Basic salts

Basic salt = Cation + **hydroxy**-anion

for example:

$\text{Cu}_2(\text{OH})_2\text{CO}_3$ Dicopper(II) di**hydroxy**carbonate

$\text{Ca}(\text{OH})\text{Cl}$ Calcium **hydroxy**chloride

$\text{Mg}(\text{OH})\text{PO}_4$ Magnesium **hydroxy**phosphate

(4).复盐(Mixed salts)：同正盐的读法。

Mixed salt = Cation + cation' + anion

for example:

NaKSO_3 Sodium potassium sulfite

CaNH_4PO_4 Calcium ammonium phosphate

AgLiCO_3 Silver lithium carbonate

NaNH_4SO_4 Sodium ammonium sulfate

KNaCO_3 : potassium sodium carbonate

$\text{NaNH}_4\text{HPO}_4$: sodium ammonium hydrogenphosphate

5) 水合盐: 结晶水读做water或hydrate

如 $\text{AlCl}_3 \cdot 6\text{H}_2\text{O}$: aluminum chloride 6-water

或 aluminum chloride hexahydrate

$\text{AlK}(\text{SO}_4)_2 \cdot 12\text{H}_2\text{O}$: aluminium potassium sulfate 12-water

5. Acids

(1). Per-, hydro-,normal acid (its salt-ate,-ide)

Acid = Central element's root -ic + acid

for example:



Carbonic acid



Sulfuric acid



Phosphoric acid



Nitric acid



Perchloric acid



Hydrochloric acid

(2). Meta- and hypo-acid (its salt-ite)

Acid = Central element's root **-ous** + acid

for example:



Sulfurous acid



Phosphorous acid



Nitrous acid



Hypochlorous acid



Chlorous acid

含氧酸与含氧酸根阴离子

采用前后缀的不同组合显示不同价态的含氧酸和含氧酸根阴离子，价态相同的含氧酸及含氧酸根阴离子具有相同的前缀，不同的后缀。

高某酸	per-ic	正酸	-ic	亚酸	-ous	次酸	hypo-ous
高某酸根	per-ate	正酸根	-ate	亚酸根	-ite	次酸根	hypo-ite

其它的前缀还有 ortho-正 meta-偏 thio-硫代

举例：	HClO_4	perchloric acid	ClO_4^-	perchlorate ion
	HClO_3	chloric acid	ClO_3^-	chlorate ion
	HClO_2	chlorous acid	ClO_2^-	chlorite ion
	HClO	hypochlorous acid	ClO^-	hypochlorite ion
	H_2SO_4	sulfuric acid	H_2SO_3	sulfurous acid
	HNO_3	nitric acid	HNO_2	nitrous acid
	HPO_3	metaphosphoric acid	$\text{S}_2\text{O}_3^{2-}$	thiosulfate ion

络合物的命名 (Naming coordination complex)

命名时先命名阳离子部分，最后命名阴离子部分，阴离子配体以字母顺序列出，中心阳离子价态一般以罗马数字在名称后标出。

$K[BF_4]$ potassium tetrafluoroborate(III)

$K_4[Fe(CN)_6]$ potassium hexacyanoferrate(II)

$[Cu(NH_3)_4]SO_4$ Tetraamminecopper(II) sulfate

$[Co(H_2O)_2(NH_3)_2(CO_2)]NO_3$

Diammine diaquacarbonatocobalt(III)nitrate

Naming coordination complex

1. Ligands

(1). Negative ions as ligands

Ligand = Element's root **-O**

for example:

CN^- Cyano

F^- Fluoro

Cl^- Chloro

Br^- Bromo

O^{2-} Oxo

OH^- Hydroxo

NO_2^-

Nitro

NO_3^-

Nitrato

CO_3^{2-}

Carbonato

CH_3COO^-

Acetato

H^-

Hydrido

$^{-}\text{O}_2\text{CCO}_2^{-}$

Oxalato

(2). Neutral molecules as ligand

Ligand = Radical name

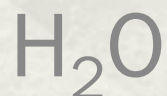
for example:



Ammine



Carbonyl



Aqua



Methylamine



Ethylenediamine

2. Complex ions

(1). Neutral complex or complex ions with positive charge

Complex ion = n-Ligand-metal ion(N)

for example:

$\text{Ag}(\text{NH}_3)_2^+$ Diamminesilver(I)

$\text{Cu}(\text{NH}_3)_4^{2+}$ Tetraamminecopper(II)

$[\text{Co}(\text{NH}_3)_3(\text{NO}_2)_3]$ Triamminetrinitrocobalt(III)

常见配体的名称

ions		molecules	
CN ⁻	cyano	H ₂ O	aqua
OH ⁻	hydroxo	NH ₃	ammine
CH ₃ COO ⁻	acetato	CO	carbonyl
NO ₃ ⁻	nitrato	CH ₃ NH ₂	methylamine
NO ₂ ⁻	nitro		
F ⁻	fluoro		
Cl ⁻	chloro		
Br ⁻	bromo		
CO ₃ ²⁻	carbonato		

(2).Complex ions with negative charge

Complex ion = n-Ligand-metal's root-ate(N)

for example:

$[\text{Fe}(\text{CN})_6]^{4-}$ Hexafluoroferrate(II)

$[\text{BF}_4]^-$ Tetrafluoroborate(III)

$[\text{AlF}_6]^{3-}$ Hexafluoroaluminate(III)

$[\text{AuCl}_4]^-$ Tetrachloroaurate(III)

3.Naming complex

Complex = Cation + anion

for example:

$\text{Li}[\text{AlH}_4]$ Lithium tetrahydroaluminate(III)

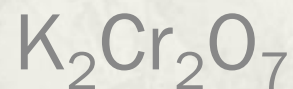
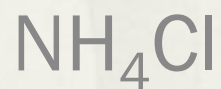
$[\text{Ag}(\text{NH}_3)_2]\text{Cl}$ Diamminesilver(I) chloride

$\text{K}_4[\text{Fe}(\text{CN})_6]$ Potassium hexacyanoferrate(II)

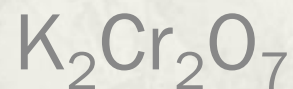
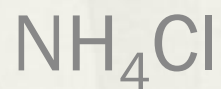
$[\text{Cu}(\text{NH}_3)_4]\text{SO}_4$ Tetraamminecopper(II) sulfate

$\text{Ni}(\text{CO})_4$ Tetracarbonylnickel(0)

Exercise



Exercise



Answer

- * H_2SO_4 sulfuric acid
- * HCl hydrogen chloride or hydrochloric acid
- * HNO_3 nitric acid
- * HNO_2 nitrous acid
- * HCN hydrogen cyanide or hydrocyanic acid
- * Na_2S sodium sulfide
- * CuSO_4 copper (II) sulfate or cupric sulfate
- * $\text{Fe}(\text{NO}_3)_3$ iron (III) nitrate or ferric nitrate
- * HClO_4 perchloric acid
- * KCN potassium cyanide
- * NH_4Cl ammonium chloride

Answer

- * NaClO sodium hypochlorite
- * NaOH sodium hydroxide
- * Mn(OH)_2 Manganese(II) hydroxide
- * Fe_2O_3 iron(III) oxide or ferric oxide
- * P_2O_5 Diphosphorus pentoxide
- * H_2O_2 hydrogen peroxide
- * $\text{K}_2\text{Cr}_2\text{O}_7$ potassium dichromate
- * $\text{Cu}_2(\text{OH})_2\text{CO}_3$ Dicopper(II) dihydroxycarbonate
- * CaHPO_4 calcium hydrogen phosphate
- * PtCl_4^{2-} tetrachloroplatinum(II)
- * $[\text{Ag}(\text{NH}_3)_2]\text{Cl}$ Diamminesilver(I) chloride
- * $\text{K}_4[\text{Fe}(\text{CN})_6]$ Potassium hexacyanoferrate(II)

二 有机物的命名

1 烷烃 (alkanes)

1.1 直链烷烃 烃类化合物的命名是有机命名的基础。英文名称除了含1到4个碳原子以外，其余均用希腊文和拉丁文的数词加上相应的词尾 (-ane) 来命名，10个碳原子以上的则在数词前加前缀un、do、tri、tetra、penta等。

如：甲烷 methane 乙烷 ethane 丙烷 propane
丁烷 butane 戊烷 pentane 己烷 hexane
庚烷 heptane 辛烷 octane 壬烷 nonane
癸烷 decane 十一烷 undecane 十二烷 dodecane

Alkane = Number prefix-**ane**

for example:

CH_4 Methane $\text{CH}_3\text{CH}_2\text{CH}_3$ Propane

CH_3CH_3 Ethane $\text{CH}_3(\text{CH}_2)_2\text{CH}_3$ Butane

$\text{CH}_3(\text{CH}_2)_3\text{CH}_3$ Pentane $\text{CH}_3(\text{CH}_2)_4\text{CH}_3$ Hexane

$\text{CH}_3(\text{CH}_2)_5\text{CH}_3$ Heptane $\text{CH}_3(\text{CH}_2)_6\text{CH}_3$ Octane

$\text{CH}_3(\text{CH}_2)_7\text{CH}_3$ Nonane $\text{CH}_3(\text{CH}_2)_8\text{CH}_3$ Decane

11~19Alkane = Number prefix-decane

for example:

11-alkane

Undecane

12-alkane

Dodecane

13-alkane

Tridecane

14-alkane

Tetradecane

15-alkane

Pentadecane

16-alkane

Hexadecane

17-alkane

Heptadecane

18-alkane

Octadecane

19-alkane

Nonadecane

20-alkane

Icosane

21~29 Alkane = Number prefix-cosane

for example:

21-alkane

Henicosane

22-alkane

Docosane

23-alkane

Tricosane

24-alkane

Tetracosane

25-alkane

Pentacosane

30-Alkane

triacontane

31~39Alkane = Number prefix-**triacontane**

for example:

31-Alkane

Hentriacontane

32-Alkane

Dotriacontane

33-Alkane

Tritriacontane

34-Alkane

Tetratriacontane

35-Alkane

pentatriacontane

36-Alkane

Hexatriacontane

40~90Alkane = Number prefix-**contane**

for example:

40 Alkane

Tetracontane

50 Alkane

Pentacontane

60 Alkane

Hexacontane

70 Alkane

Heptacontane

80 Alkane

Octacontane

90 Alkane

Nonacontane

100 Alkane

Hectane

1.2 含支链烷烃和烷基

命名含支链的烷烃时，可把它们视为直链烷烃，但分别是某些氢（hydrogen）原子被称为烷基（alkyl groups）的原子取代。命名烷基时，只需把“基”（-yl）字加在相应的烷烃的字首后。

如： CH_3 - Methyl $\text{CH}_3-(\text{CH}_2)_9-\text{CH}_2$ - Undecyl
 $\text{CH}_3-\text{CH}_2-\text{CH}(\text{CH}_3)-\text{CH}_2-\text{CH}_3$ 3-Methylpentane

Univalent radicals

Radical = Alk -yl

for example:



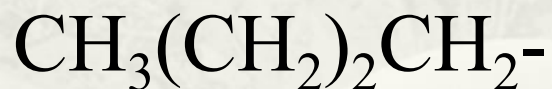
Methyl



Ethyl



Propyl

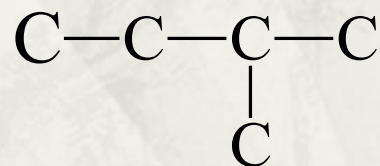


Butyl

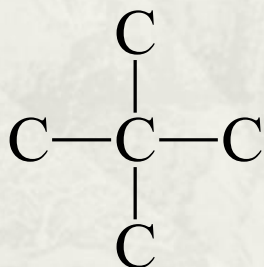
Saturated branched-chain hydrocarbon

branched-chain hydrocarbon =
n-Radical+alkane

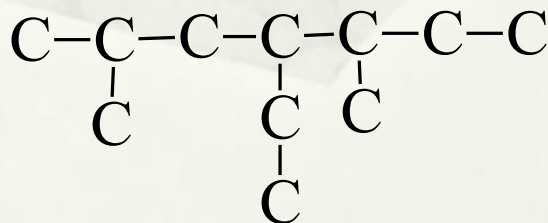
for example:



2-Methylbutane



2,2-Dimethylpropane

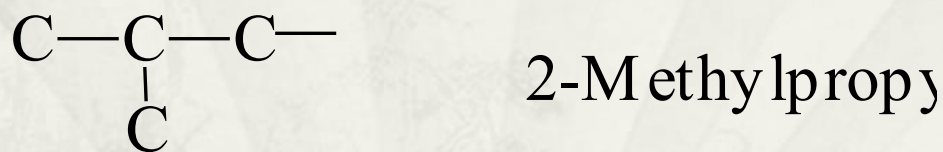
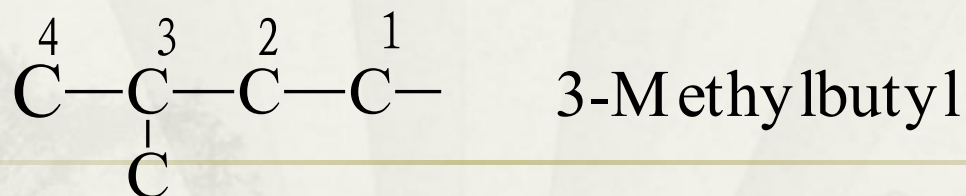


4-Ethyl-2,5-dimethylheptan

Univalent branched radicals

Radical = Alk -yl

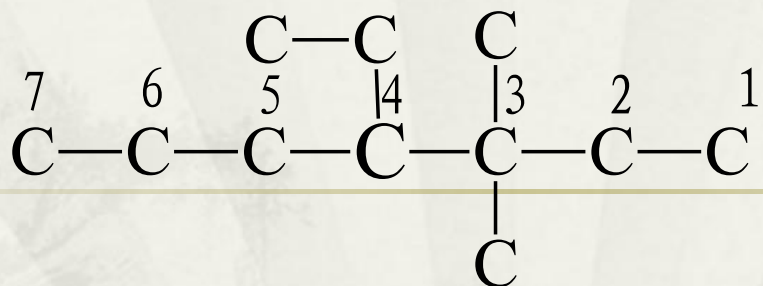
for example:



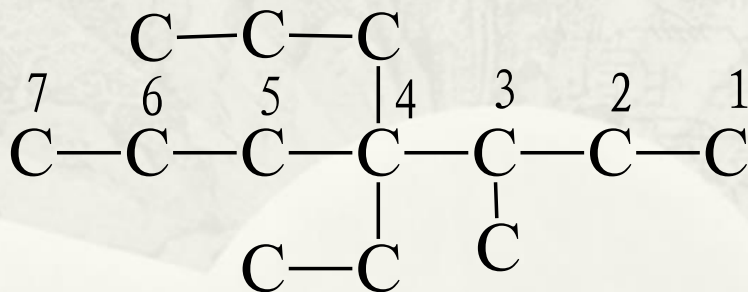
Polyside chain hydrocarbon

Butyl > Ethyl > Isopropyl > Methyl > Propyl

for example:



4-Ethyl-3,3-dimethylheptane



4-Ethyl-3-methyl-4-propylheptane

1.3 一些可采用普通命名的支链烷烃：

$(\text{CH}_3)_2\text{CH}-\text{CH}_3$	Isobutane
$(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}_3$	Isopentane
$(\text{CH}_3)_4\text{C}$	Neopentane
$(\text{CH}_3)_2\text{CH}-\text{CH}_2-\text{CH}_2-\text{CH}_3$	Isohexane

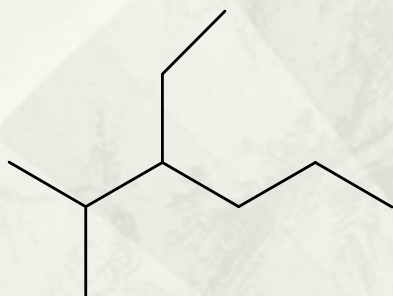
对取代的烷基也可以在相应的烃名前加“异”(iso-)、“仲”(sec-)、“叔”(tert-)、“新”(neo-)等字命名。下面是一些符合条件的烷基，其系统命名和普通命名如下：

$(\text{CH}_3)_2\text{CH}-\text{CH}-$	2-Methylpropyl	Isobutyl
$\text{CH}_3-\text{CH}_2-\text{CH}(\text{CH}_3)-$	1-Methylpropyl	sec-Butyl
$(\text{CH}_3)_3\text{C}-$	1,1-Dimethylethyl	tert-Butyl
$\text{CH}_3-\text{CH}_2-\text{C}(\text{CH}_3)_2-$	1,1-Dimethylpropyl	tert-pentyl
$(\text{CH}_3)_3\text{C}-\text{CH}_2-$	2,2-Dimethylpropyl	Neopentyl

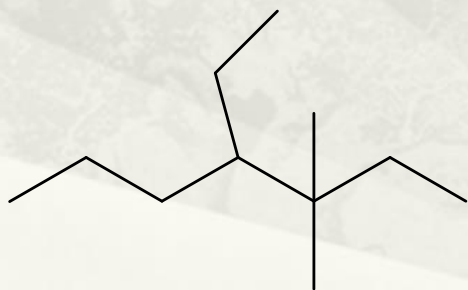
1.4 复杂的烷烃

结构较复杂的烷烃不能用普通命名法命名，只能采用系统命名法。选最长的碳链为主链，按相应的直链烷烃命名，从一端向另一端编号，支链作为取代基放在母体名称前，编号时使支链的编号尽可能小且支链的排列按基团的字母顺序。

如:



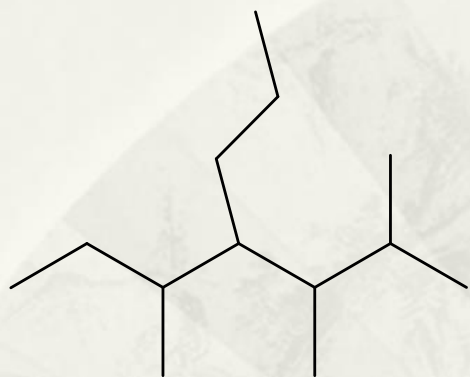
3-Ethyl-2-methylhexane



4-ethyl-3,3-dimethylheptane

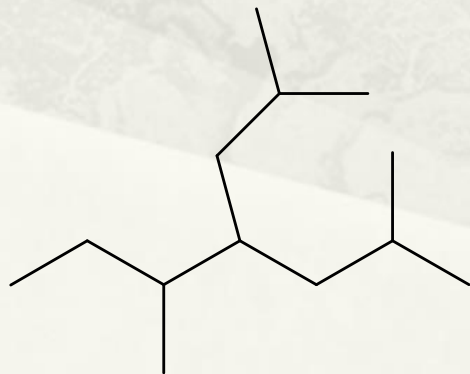
复杂的烷烃命名时须注意分子中有两个等长碳链时，按以下原则进行比较：

1) 带支链数目较多者为主链，例如：



2,3,5-Trimethyl-4-propylheptane

2) 支链定位号较小者为主链

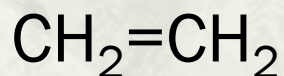


4-Isobutyl-2,5-dimethylheptane

2 烯烃和炔烃 (alkenes and alkynes)

2.1 命名

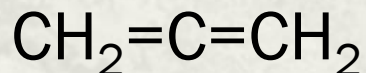
烯烃和炔烃命名时将相应的烷烃的词尾“烷”(ane)改为“烯”(ene)或“炔”(yne)，名称前加上不饱和键的编号即可。当所带的双键或叁键不止一个时，可在前边加上di、tri、tetra等数字来表示。有些简单的烯炔类化合物可用普通名称。例如：



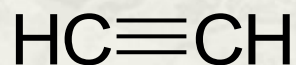
Ethene



1-Pentene



1,2-Propadiene



Ethyne



1,3-Butadiyne

Naming unsaturated hydrocarbon

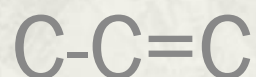
2.1 Unbranched acyclic alkene

Alkene = Number prefix-ene

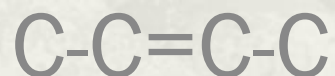
for example:



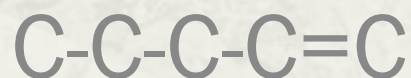
2-Hexene



Propene



2-Butene



1-Pentene

Poly-ene

n-Alkene = Alk-a-n-ene

n=2

-adiene

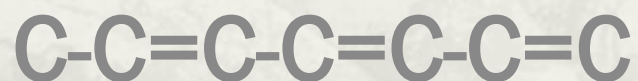
n=3

-atriene

n=4

-atetraene

for example:



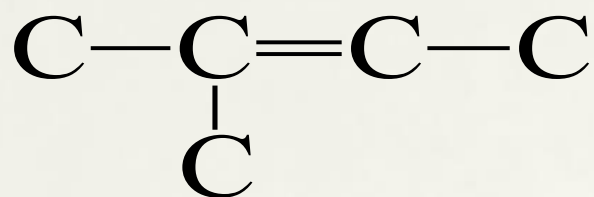
1,3,5-Heptatriene



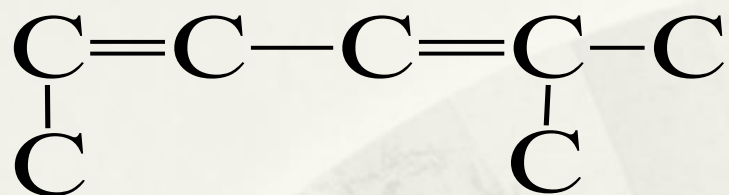
1,3-Butadiene



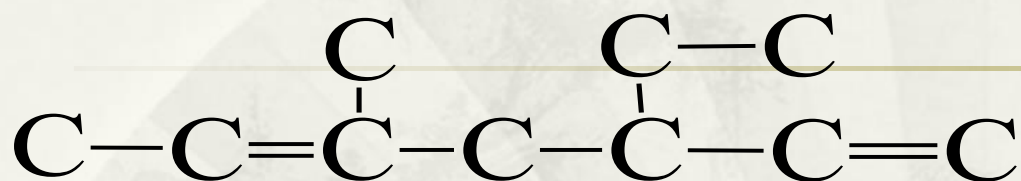
1,3-Pentadiene



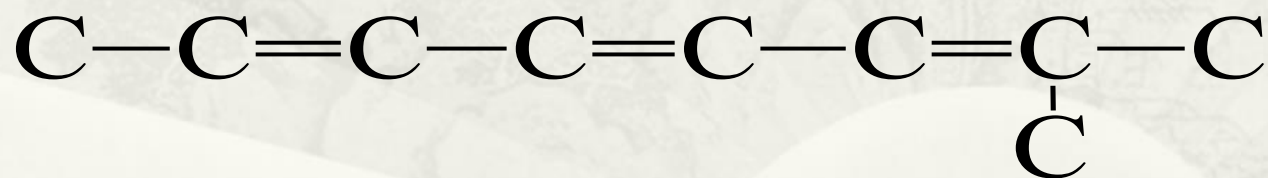
2-Methyl-2-butene



2-Methyl-2,4-hexadiene



3-Ethyl-4-Methyl-1,5-heptadiene



2-Methyl-2,4,6-octatriene

Alkyne

Alkyne = Alk-yne (-a-n-yne)

n=2

-adiyne

n=3

-atriyne

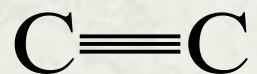
for example:



Propyne



2-Pentyne



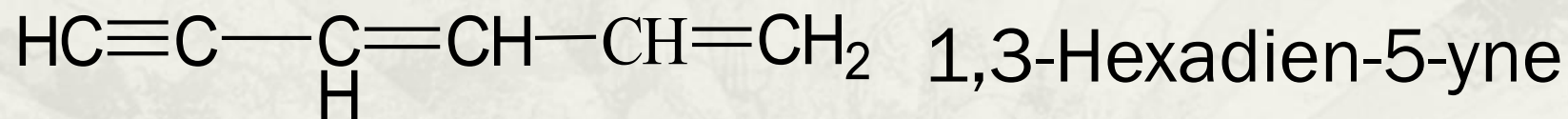
Ethyne (acetylene)



1,3-Butadiyne

2.2 烯炔的命名

烃类分子中同时含有双键和叁键时成为烯炔，命名时烯在前炔在后，双键的编号写在前面，叁键的定位号写在表示炔烃词尾之前。例如：



Unsaturated hydrocarbon with both double and triple bonds

Alkenyne = Number prefix-en-yne

Two ene + one yne

-adien-?-yne

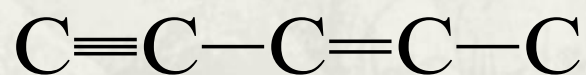
Three ene + one yne

-atrien-?-yne

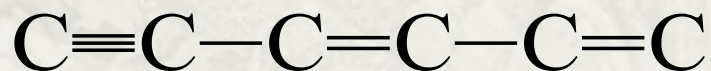
One ene + two yne

-en-?-diyne

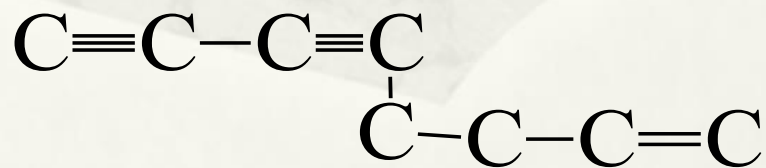
for example:



3-Penten-1-yne



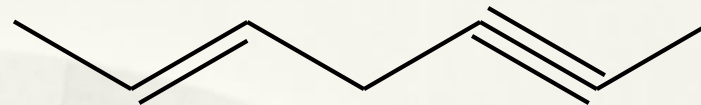
1,3-Hexadien-5-yne



7-Octene-1,3-diyn



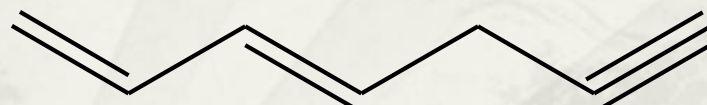
1-penten-4-yne



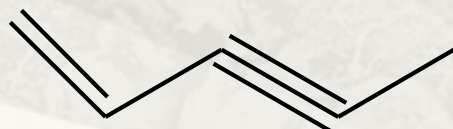
2-hepten-5-yne



4-hexen-1-yne

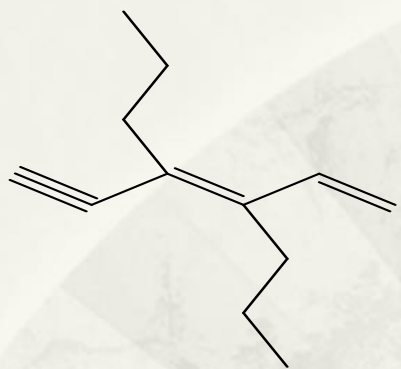


1,3-heptadien-6-yne

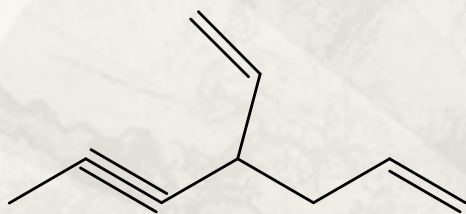


1-hexen-4-yne

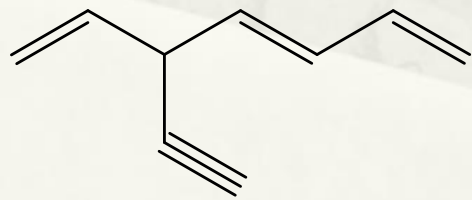
系统命名中，应选含不饱和键最多且最长的直链。如分子中有两条直链具有相同数目的不饱和键时，取碳原子数较多者为主链。如碳原子数相同，则取含双键数目较多者为主链。例如：



3,4-Dipropyl-1,3-hexadien-5-yne



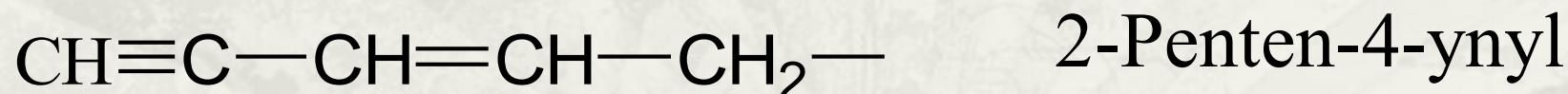
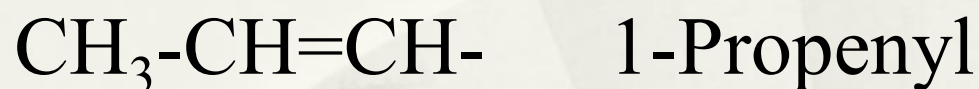
4-Vinyl-1-hepten-5-yne



5-Ethynyl-1,3,6-heptatriene

2.3 不饱和烃基的命名

不饱和烃基的命名类似于饱和烃基，但需要标出不饱和键的位置。有些简单的不饱和烃基可用俗名。例如：



Alkene radical=Alken-**yl**

Alkyne radical=Alkyn-**yl**

Alkene radical=Alken-yl

Alkyne radical=Alkyn-yl

for example:

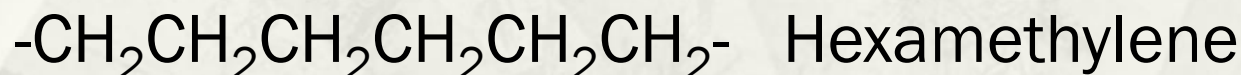
$\text{C}\equiv\text{C}-$	Ethynyl
$\text{C}\equiv\text{C}-\text{C}-$	2-Propynyl
$\text{C}-\text{C}=\text{C}-$	1-Propenyl
$\text{C}=\text{C}-\text{C}=\text{C}-$	1,3-Butadienyl
$\text{C}\equiv\text{C}-\text{C}=\text{C}-\text{C}-$	2-Penten-4-ynyl

2.4 多价基的命名

1) 两个自由价在同一个碳原子上称为“亚”，英文词尾为“-ylidene”。例如：



2) 两个自由价不在同一个碳原子上也称为“亚”，但需要标出定位号。例如：



3) 三个自由价在同一个碳原子上的称为次基，英文词尾为-ylidyne。例如：



Multivalent radicals

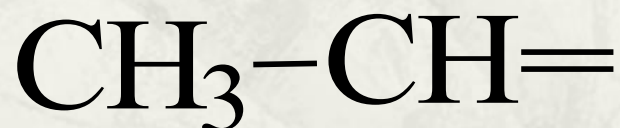
Bivalent radical = univalent radical-idene (ene)

Trivalent radical = univalent radical-idyne

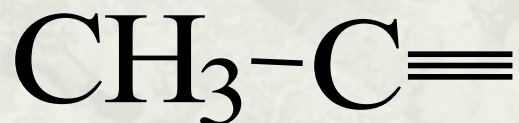
for example:



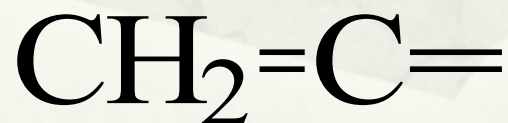
Methylidyne



Ethylidene



Ethylidyne



Vinylidene



Methylene



Ethylene



Trimethylene



Propylene



Tetramethylene

3 环烃(cyclic hydrocarbons)

3.1 脂肪环烃(alicyclic hydrocarbons)

3.1.1 未取代的饱和单环烃命名时刻在相应的开链烃名前加“环”（cyclo）字。例如：



Cyclopropane



cyclobutane



Cyclohexane



cyclopentane

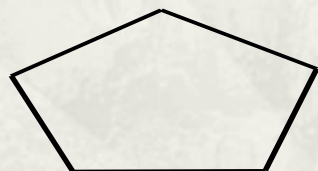
Saturated monocyclic hydrocarbons

Cyclic alkane = Cyclo- alkane

for example:



Cyclopropane



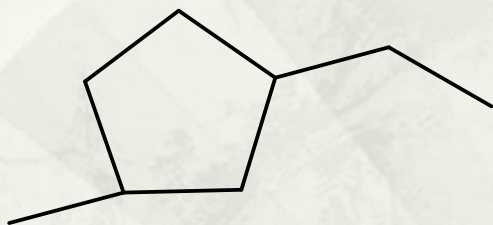
Cyclopentane



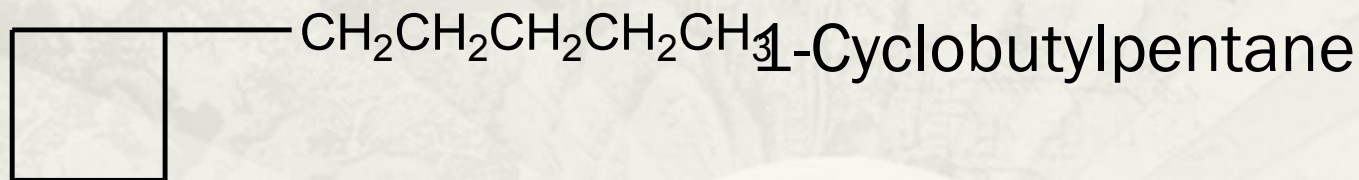
Cyclohexane

3.1.2 含取代基的脂肪环烃

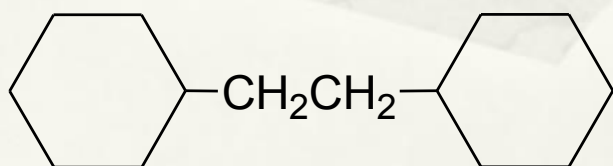
环上带有侧链时，如侧链的碳原子数不比环内的碳原子数多，侧链作为取代基。如侧链的碳原子数目等于或多于环内碳原子数，或侧链有不只一个脂环时，则将环作为取代基命名。例如：



1-Ethyl-3-methylcyclopentane



1-Cyclobutylpentane

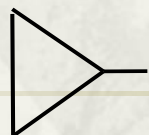


1,2-Dicyclohexylethane

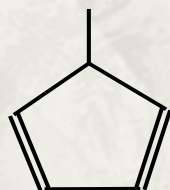
Univalent radicals of cyclic hydrocarbon

Cyclic radical = Cyclo-R

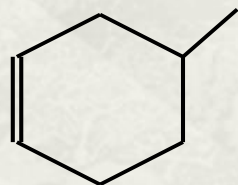
for example:



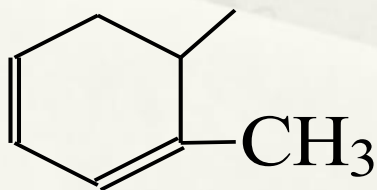
2-Cyclopropyl



2,4-Cyclopentadien-1-yl



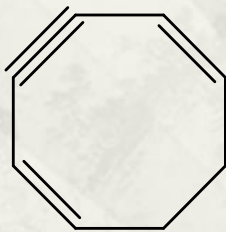
3-Cyclohexen-1-yl



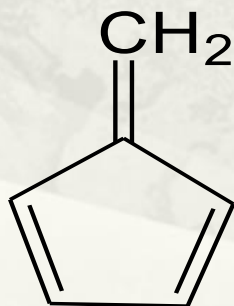
2-Methyl-2,4-cyclohexadien-1-y

3.1.3 不饱和单环烃的命名

不饱和单环烃的命名是把相应的饱和单环烃的词尾ane改为ene（烯）、yne（炔）、adiene（二烯）、adiyne（二炔）、enyne（烯炔）等，并使不饱和键尽可能取最小编号。例如：



1,5-Cyclooctadien-3-yne

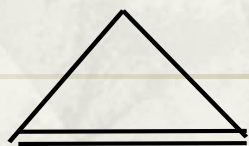


5-Methylene-1,3-cyclopentadiene

Unsaturated monocyclic hydrocarbons

Cyclic alkene = Cyclo-alkene

for example:



Cyclopropene



Cyclopentadiene



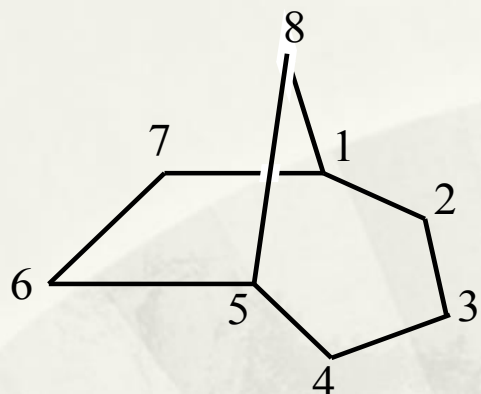
1-Cyclohexen-4-yne

4 桥环烃(bridgedhydrocarbons)

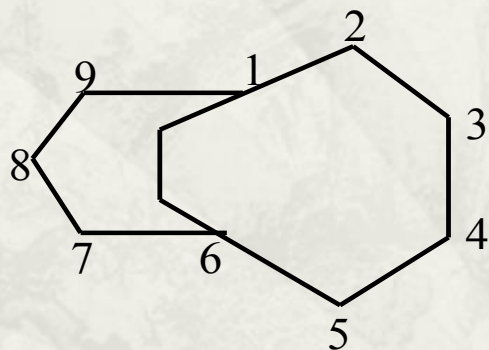
桥环烃的命名按以下步骤进行

- 1) 确定环数，将环状分子变为开链烃时所需要断开的次数就是该分子的环数。
- 2) 选择主环，主环应包含尽量多的碳原子，且有两个碳原子作为主桥桥头。
- 3) 选择主桥，主桥要包含尽量多的碳原子，且尽可能把主环对称的分开。
- 4) 桥环烃的编号 桥头原子定为1，然后沿最长链依次编号至另一桥头，再沿次长链编至1号碳原子，再编主桥和次桥内的碳原子直至编完。
- 5) 将相当于桥环烃所有环上碳原子总数的直链烃名放在后面。
- 6) 在最前面，标出环数，如双环Bicyclo,三环Tricyclo。然后在方括号内标明各个环中的除桥头原子以外的碳原子数目，最后标明主桥及次桥，并用上标来表明次桥的位置，标明次桥位置的上应尽可能小。

Bicyclic bridged hydrocarbon



Bicyclo[3,2,1]octane

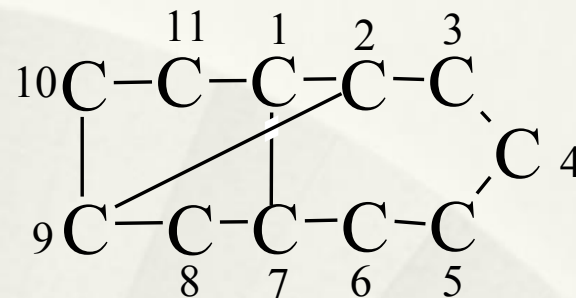
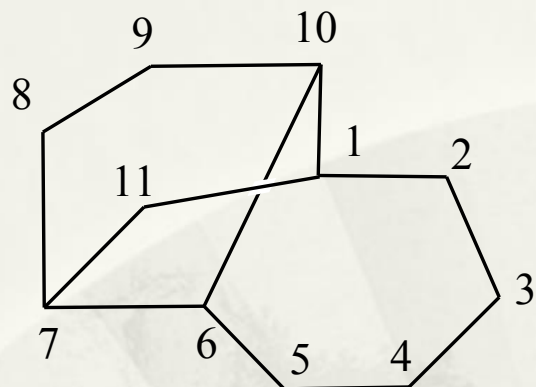


Bicyclo[4,3,2]undecane

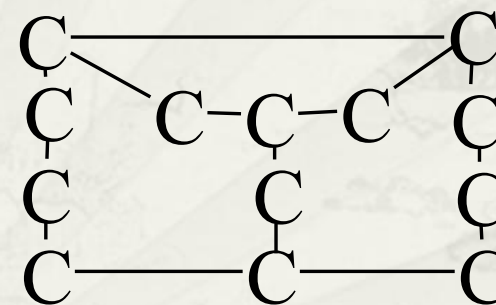
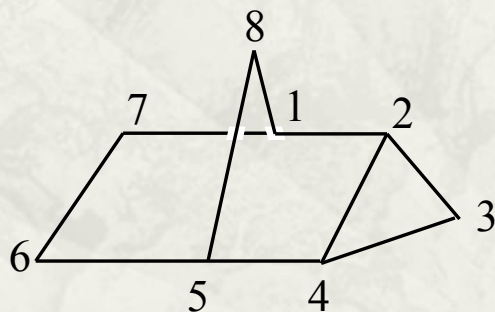


Bicyclo[2,2,0]hexane

Polycyclic systems



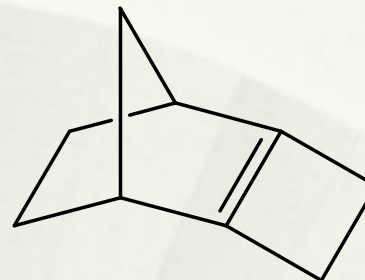
Tricyclo[5,3,1,0^{6,10}]undecane Tricyclo[5,4,0,0^{2,9}]undecane



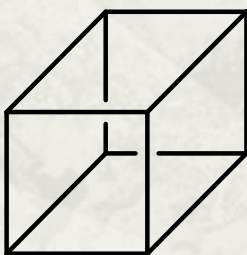
Tricyclo[3,2,1,0^{2,4}]octane Tricyclo[5,5,1,0^{3,11}]tridecane



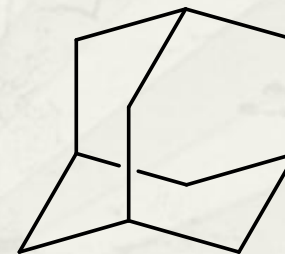
Bicyclo[4.4.0]decane



Tricyclo[4.2.1.0^{2,5}]non-2-ene



Pentacyclo[4.2.0.0^{2,5}.0^{3,8}.0^{4,7}]octane



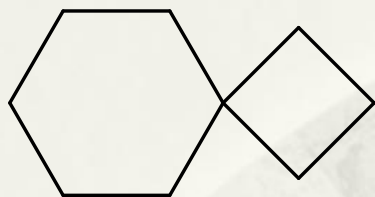
Tricyclo[3.3.1.1^{3,7}]decane

5 螺环烃(spirohydrocarbons)

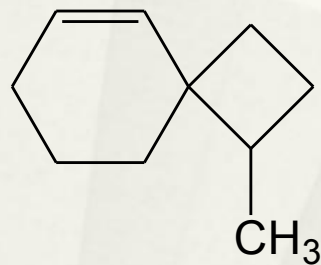
螺环烃主要采用Patterson命名法，其命名方法如下：

在与总碳原子数目相应的烃名前加螺（Spiro），中间加一方括号，将各环内螺原子以外的原子数由小到大排列在方括号内，数字之间用圆点分开，编号时从小环一端与螺原子相邻的碳原子开始沿环依次进行，待螺原子编完后再编另一环。分子中若有两个或三个螺原子时，则用二螺（Dispiro）或三螺（Trispiro）。

例如：



Spiro[3.5]nonane

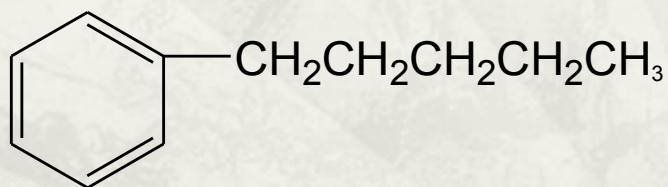


1-Methylspiro[3.5]non-5-ene

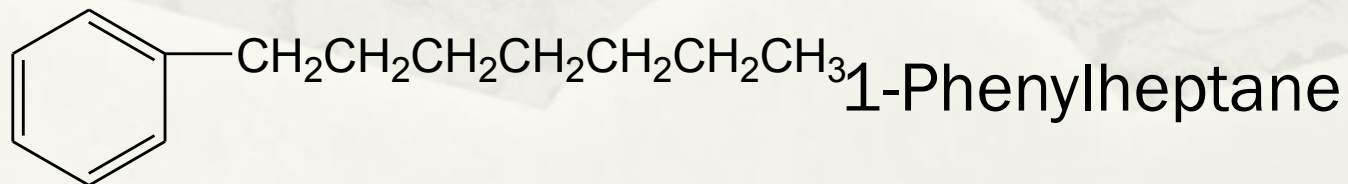
3.2 芳烃(aromatic hydrocarbons)

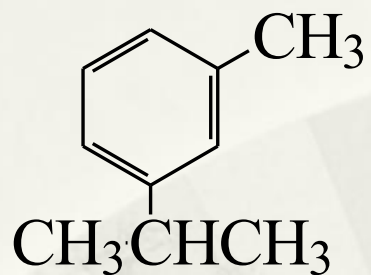
3.2.1 单环芳烃

1) 苯环上连有烃基时，苯环和烃基都可作为母体，决定于烃基的大小。两个或更多的苯环连在同一个碳原子上或碳链上时，可将苯环作为取代基命名。例如：

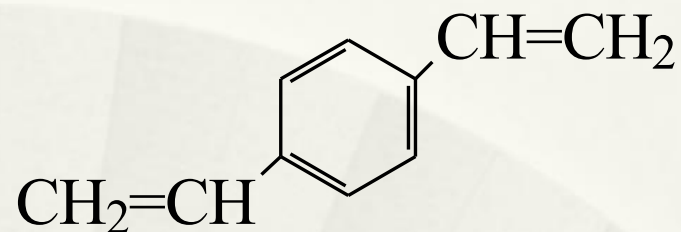


Pentylbenzene

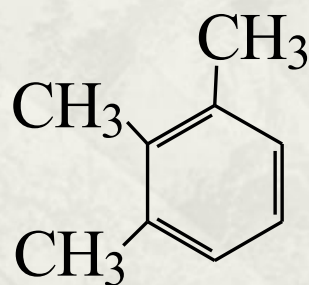




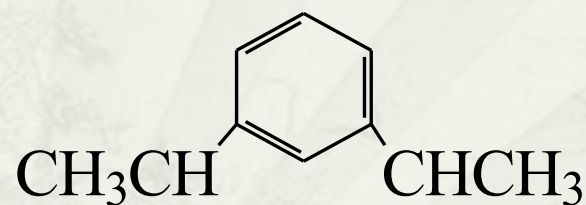
3-Isopropyl-1-Methylbenzene



1,4-Divinylbenzene

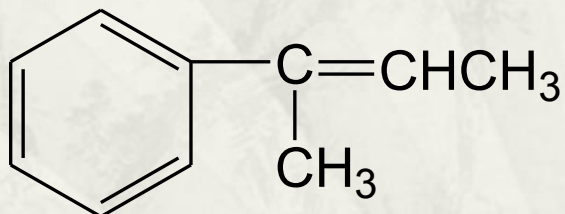


1,2,3-Trimethylbenzene

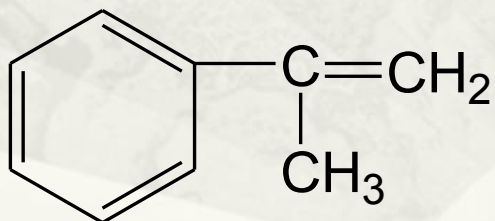


m-Diethylbenzene

2) 苯环上连有不饱和取代基时，将苯环作为链的衍生物命名，但当不饱和链不超过三个碳原子时，通常都作为苯的衍生物命名。例如：

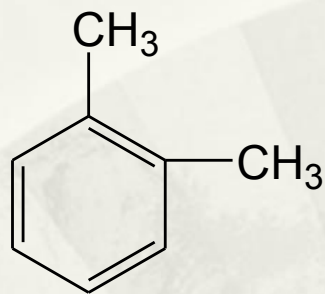


2-Phenyl-2-butene



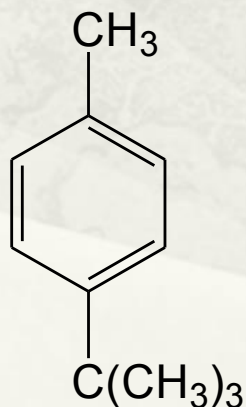
Isopropenylbenzene

3) 两个烷基取代的苯环，因为取代位置不同，可以有三个异构体，可由阿拉伯数字表示，也可用邻、间、对表示，英文名称则分别用 *o*、*m*、*p* 表示。例如：



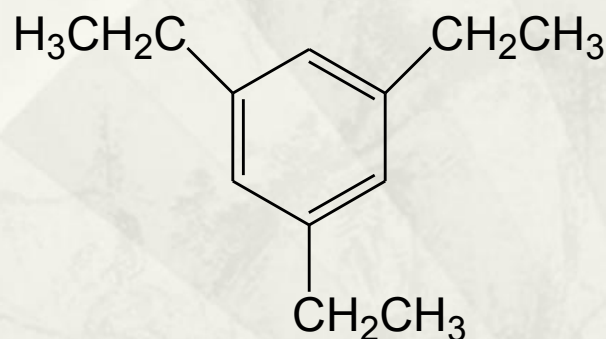
o-Dimethylbenzene

此表示方法只用于两个取代基相同或两取代基不同但其中有一个为俗名的一部分。例如：

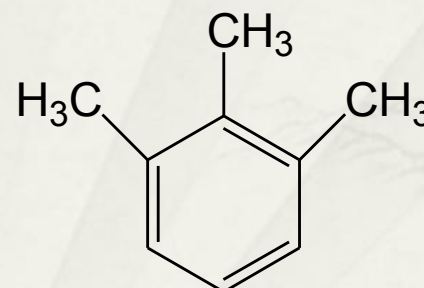


p-tert-Butyltoluene

4) 苯环上联有三个取代基时，由于它们的位置不同而常用数字定位号区别，取代基若是相同，可用“连”（vic）、“偏”（unsym）、“均”（sym）来表示。例如：

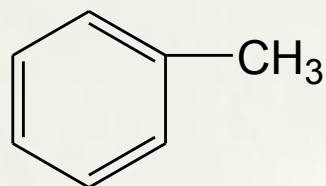


1,3,5-Triethylbenzene
或 sym-Triethylbenzene



1,2,3-Trimethylbenzene
或 vic-Trimethylbenzene

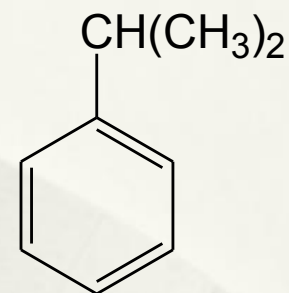
3.2.2 常用取代苯化合物的俗名 (Trivial name)



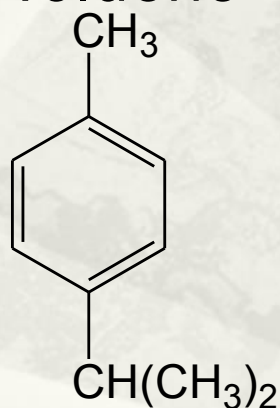
Toluene



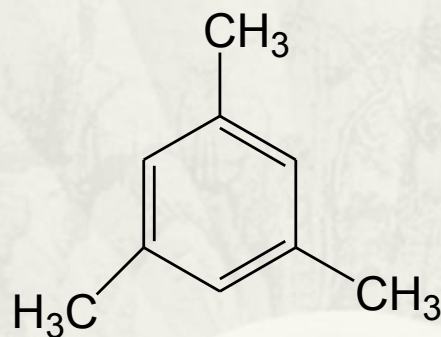
p-Xylene



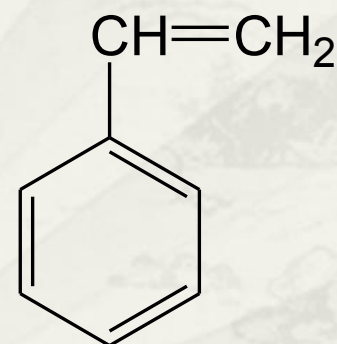
Cumene



Cymene



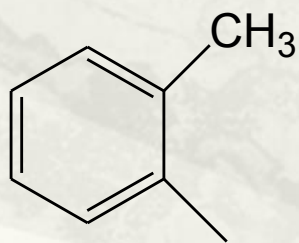
Mesitylene



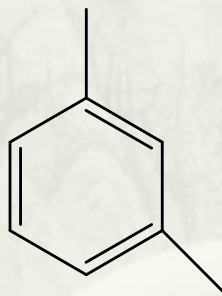
Styrene

3.2.3 芳烃基

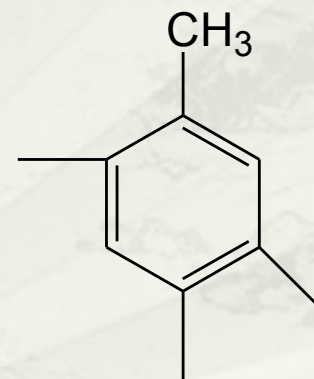
1) 从芳环上去掉一个氢原子得到的基团称为芳基 (Aryl)，去掉两个氢原子的称为亚芳基 (Arylene)。环上有取代基时，以苯基作为母体，以带自由键的碳原子为1进行编号，其它取代基的编号要尽可能小。例如：



2-Methylphenyl

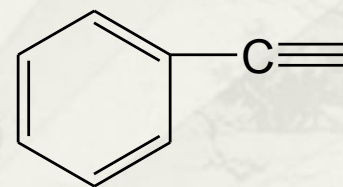
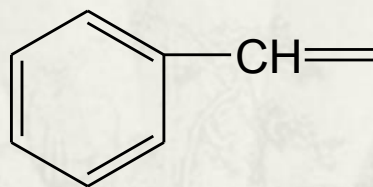
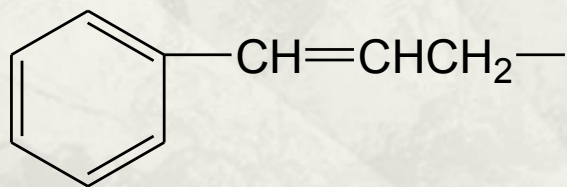


1,3-phenylene

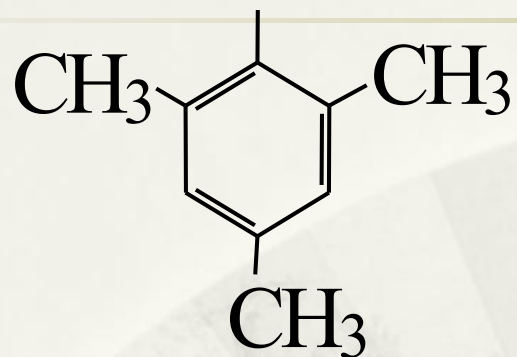


2-Methyl-1,4,5-benzenetriyl

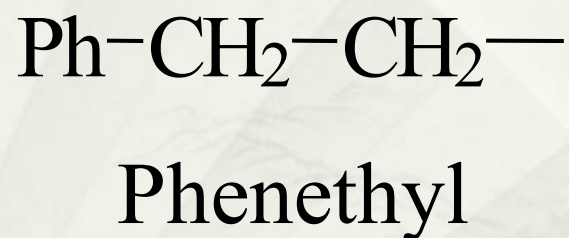
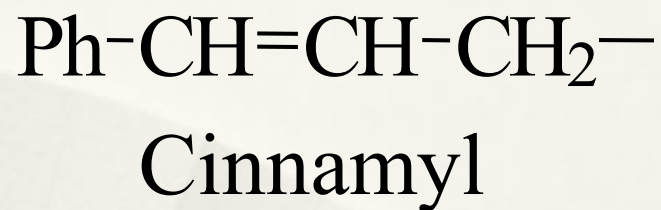
2) 单环芳烃侧链上去掉氢原子，生成的一价及多价基作为芳烃基取代的链烃基，按链烃基的原则命名。例如：



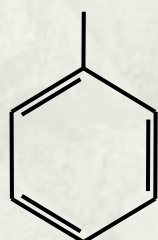
3-Phenyl-2-propenyl Phenylmethylene Phenylmethyldyne



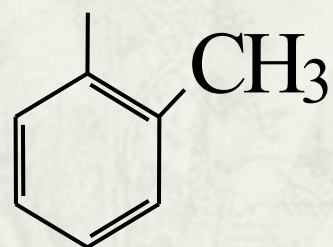
Mesityl



Benzyl



Phenyl



Tolyl



Styryl

6 醇和酚

6.1 醇(alcohols)

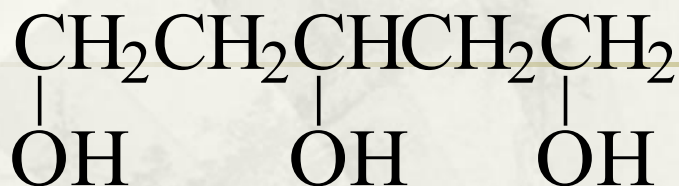
1) 系统命名法

选含羟基的最长碳链为主链，按主链烃基的碳原子数目称为某醇，英文名称是将烃基名称的词尾“e”去掉，加上“ol”。编号时从与羟基相连的碳原子开始依次向另一端编。芳醇可作为侧链醇的芳环取代物命名，此方法是主要方法。

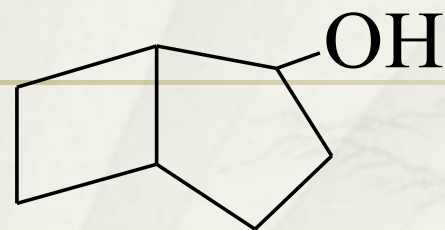
Alcohols

System: Alcohol = Alkan-ol

for example:



1,3,5-Pentanetriol



Bicyclo[3,2,0]heptan-2-ol



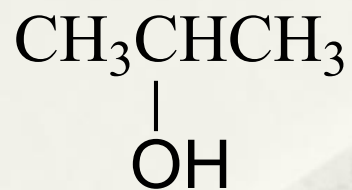
Methanol



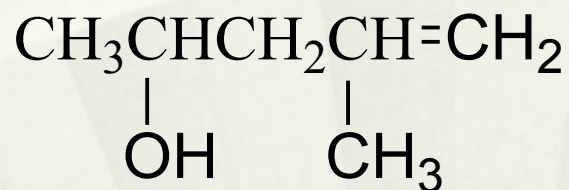
2-Propanol



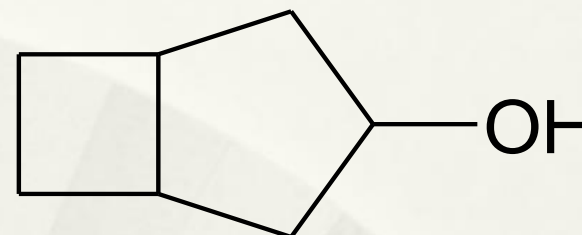
1,4-Butanediol



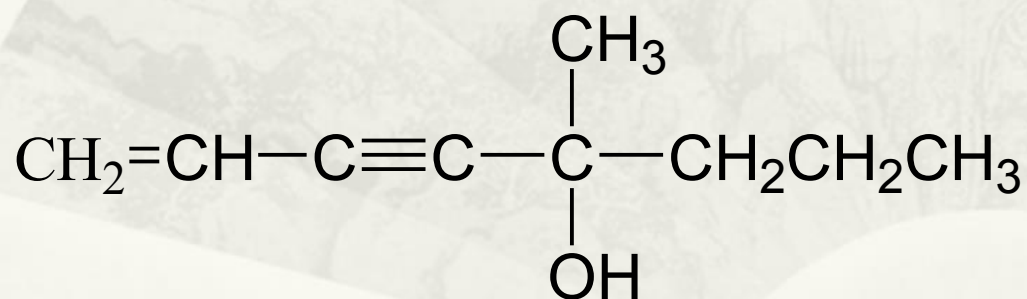
2-Propanol



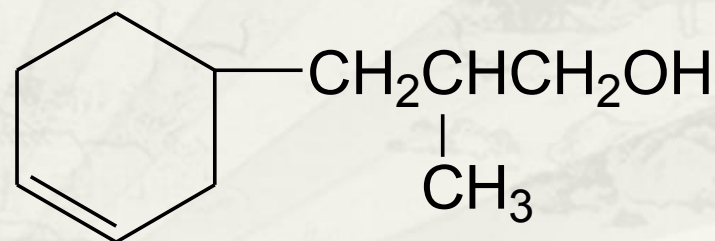
4-Methyl-4-penten-2-ol



Bicyclo[3.2.0]heptan-3-ol



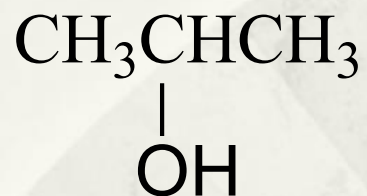
4-Methyl-7-octen-5-yn-4-ol



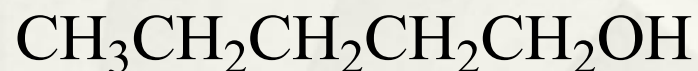
3-(3-Cyclohexen-1-yl)-2-methyl-1-propanol

2) 根基官能团命名法

在烃基后直接加“alcohol”即可，例如：



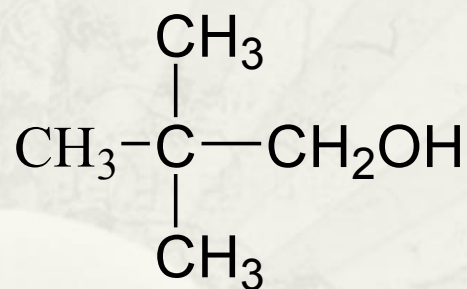
Isopropyl alcohol



Amyl alcohol



Allyl alcohol



Neopentyl alcohol

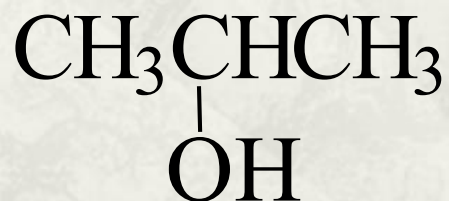
Alcohol = Radical + alcohol



Ethyl alcohol



Methyl alcohol



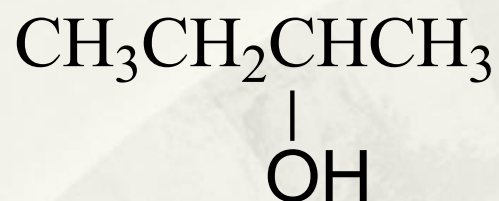
Isopropyl alcohol



Cinnamyl alcohol

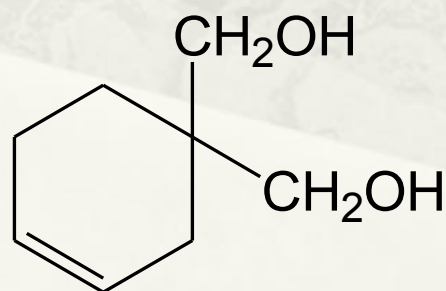
苯丙烯基

3) 衍生命名法 以甲醇作为母体，把整个分子看成是由甲醇衍生来的。例如：



Ethyl methyl methanol

4) 连接命名法 适用于羟基连在与环系相连的侧链上的醇。例如：



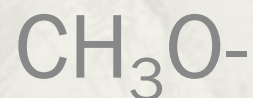
3-Cyclohexene-1,1-dimethanol

Alcohol Radicals

(1). RO- (R=C1~C4)

Radical = Alk + oxy

for example:



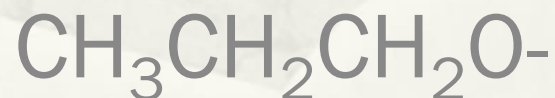
Methoxy



Ethoxy



Butoxy



Propoxy

(2). RO- (R>C4)

Radical = Alkyl-oxy

For example:



Benzyloxy



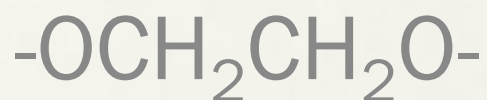
(3). -O-X-O-

Radical = X -dioxy

for example:



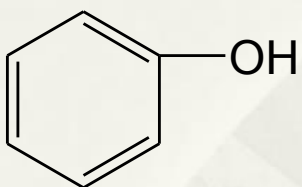
Methylenedioxy



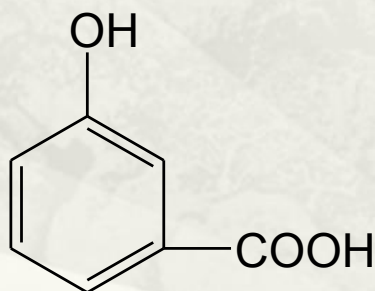
Ethylenedioxy

6.2 酚类(phenols)

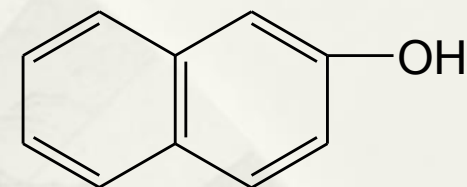
酚的命名法同醇，英文名称是把芳环系名的词尾“e”去掉，在接上词尾“ol”。分子中有更优先的官能团时，羟基作为取代基。例如：



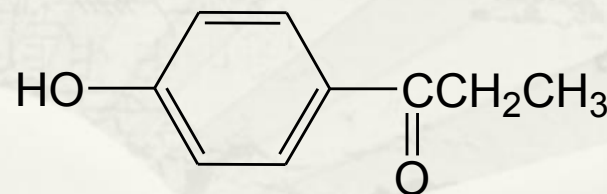
Phenol



3-Hydroxybenzenecarboxylic acid



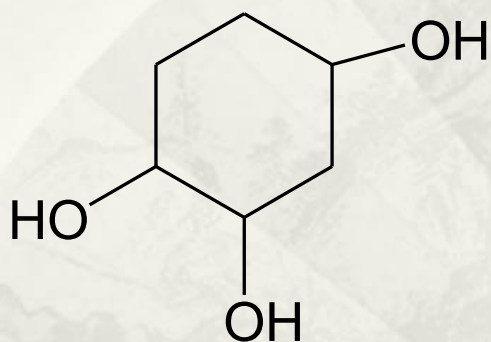
2-Naphthalenol



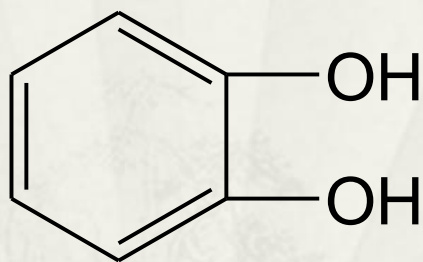
1-(4-Hydroxyphenyl)-1-propanone

6.3 多元醇和多元酚

多元醇和多元酚在烃名和“ol”间加di、tri等数词即可。例如：



1,2,4-Cyclohexanetriol



1,2-Benzenediol



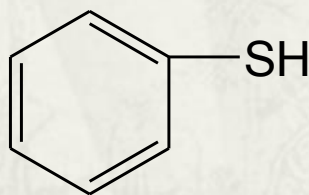
1,2,3-Propanetriol

6.4 硫醇和硫酚

其命名和相应的醇和酚一样，在醇和酚前加一“硫”（thio）字，英文名称是将“ol”改为“thiol”。作为取代基时，为“巯基”（Mercapto）。例如：



Ethanethiol



Benzenethiol



2-Mercaptoethanol

7 醚类(ethers)

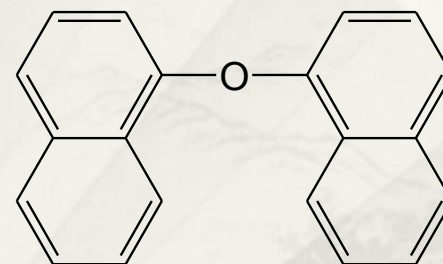
7.1 醚的命名

1) 根基官能团命名法

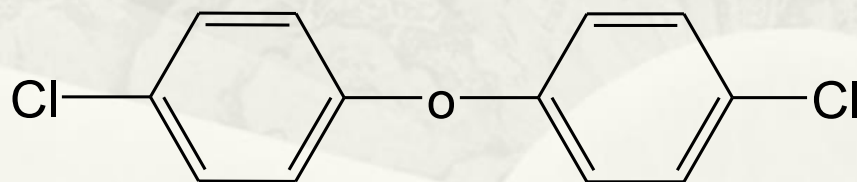
分子中两个烃基后加上“醚”(ether)即可,两个烃基相同时加“di”,若为两个复合基且相同,则加数词“bis”。例如:



Dipropyl ether



Di-2-naphthyl ether



Bis(4-Chlorophenyl) ether

Ether

Ether = R+R'+ether

for example:



Ethyl methyl ether



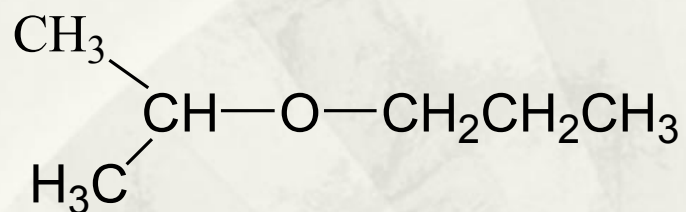
Isopropyl methyl ether



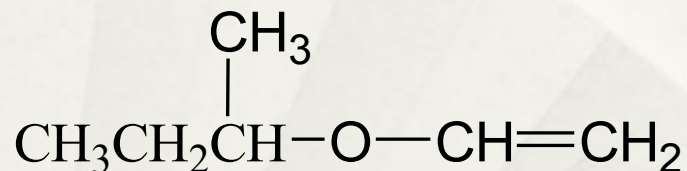
diethyl ether

2) 取代命名法

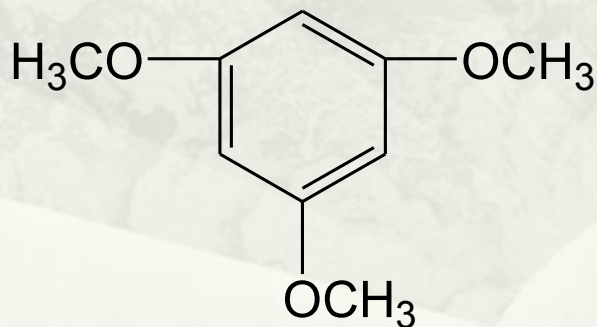
把醚分子 R_1-O-R_2 中 R_1O 作为母体 R_2 的取代基命名， RO 基为“Alcoxy”，其中 $C_1 \sim C_4$ 的分别为methoxy, ethoxy, propoxy, butoxy, C_5 以上的则称为pentyloxy, hexyloxy等。例如：



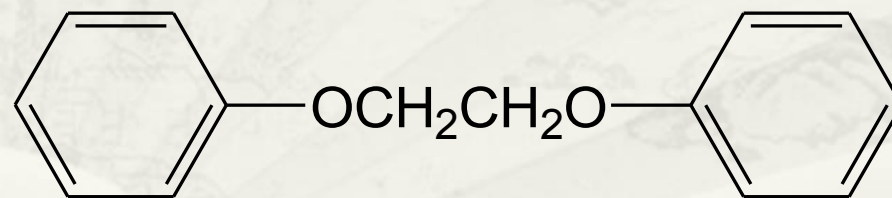
1-Isopropoxypropane



sec-Butoxyethylene



1,3,5-Trimethoxybenzene



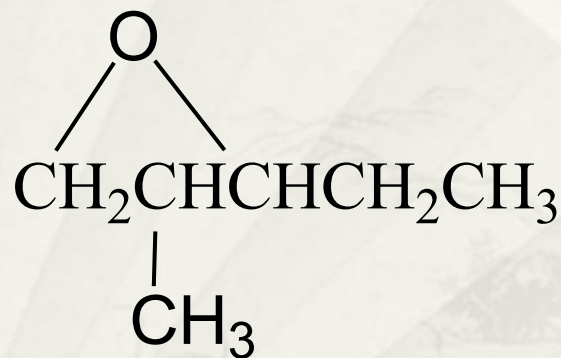
1,2-Diphenoxyethane

7.2 环氧化合物

环氧化合物用词头“环氧”（epoxy）命名。英文名中，“epoxy”可放在母体名前，不按字序排列。例如：



1-Chloro-2,3-epoxypropane
或 2,3-Epoxychloropropane



2-Methyl-1,3-epoxypentane
或 1,3-Epoxy-2-methylpentane

8 醛、酮及其衍生物

8.1 醛(aldehydes)

8.1.1 醛的命名

1) 醛的英文命名是把相应烃名的词尾“e”改为“al”。编号时将醛基的碳原子定为1号，依次向后排，含两个醛基的以两醛基间的碳链为主链，在烃名词尾“e”后加“dial”，环上直接连有二个醛基时，可在环系名后接“dicarbaldehyde”。

Aldehydes

Acyclic aldehydes

Aldehyde = Alkan -al

for example:



5-Methylhexanal



Pentanal

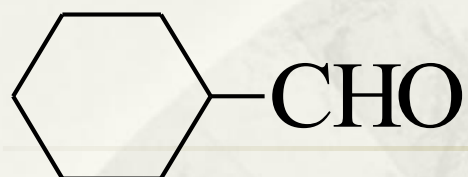


Hexanedial

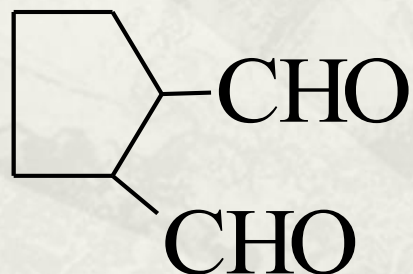
Aldehyde with cyclic group

Aldehyde = Cyclic alkane-carbaldehyde

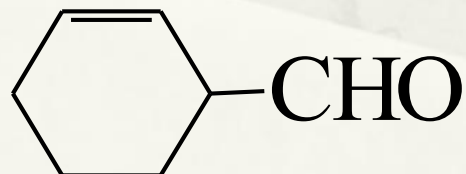
For example:



Cyclohexanecarbaldehyde



1,2-Cyclopentanedicarbaldehyde

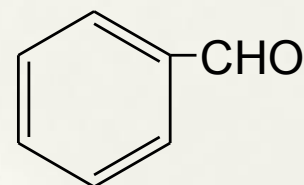


2-Cyclohexenecarbaldehyde

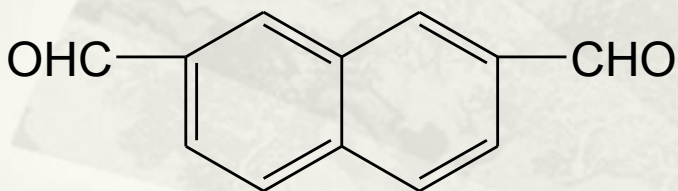
例如：



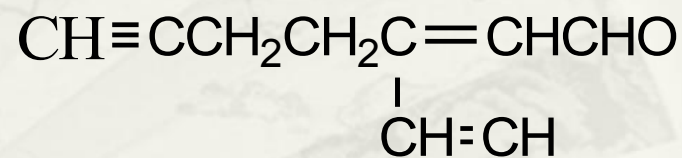
Ethanal



Benzenecarbaldehyde

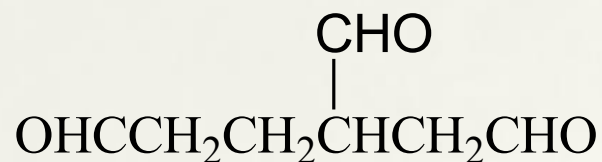


2,6-Naphthalenedicarbaldehyde

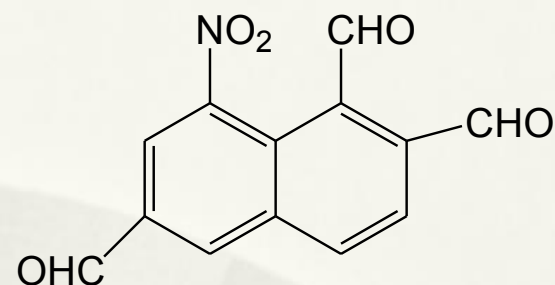


3-Vinyl-2-hepten-6-ynal

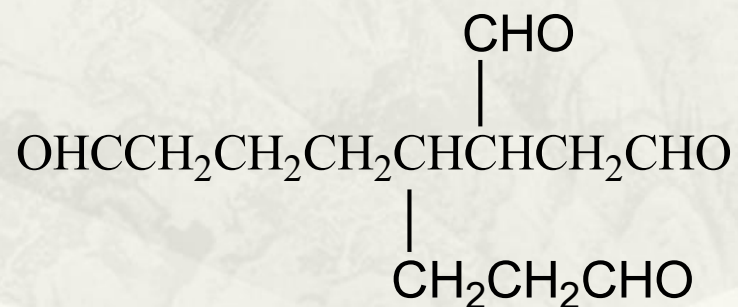
2) 含有两个以上醛基的链状多元烃，当醛基都是直链连在主链上，命名时在主链烃名后加上“三醛”或“四醛”等。英文名称在烃名后加“tricarbaldehyde”或“tetracarbaldehyde”。也可以把链中间所连的醛基作为取代基而用“Formyl”或“Methanoyl”，两端的醛基作为二醛在烃名后加“dial”。



1,2,4-Butanetricarbaldehyde
或 3-Formylhexanedial

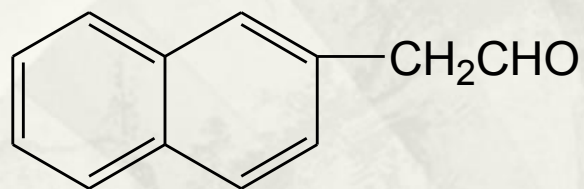


8-Nitro-1,2,6-naphthalene-tricarbaldehyde



3-(2-Formylethyl)-1,2,6-hexanetricarbaldehyde
或 3-Methanoyl-4-methanylethylhexanedial

3) 醛基通过碳链与环系连接时，可用取代命名法和连接命名法。例如：



(2-Naphthyl)acetaldehyde
Naphthaleneacetaldehyde



1,4-Phenylenediacetaldehyde or 2-

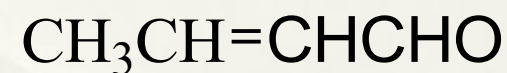
8.1.2 几种醛的俗名



Formaldehyde

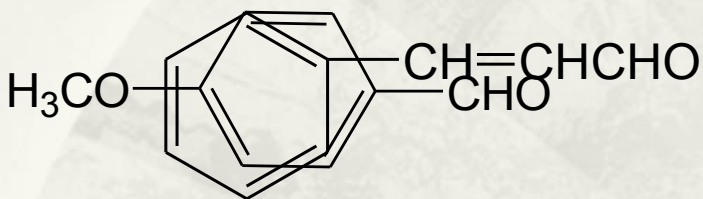
Acrylaldehyde

丙烯醛



Crotonaldehyde

巴豆醛 or 2-丁烯醛

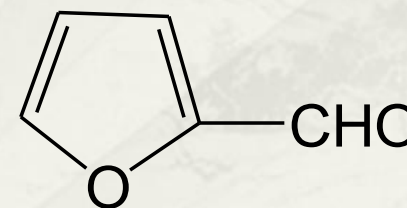


Cinnamaldehyde

肉桂醛

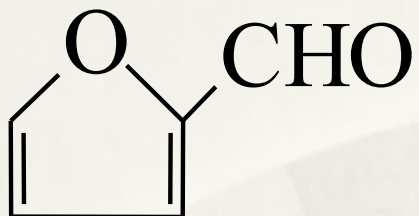
Anisaldehyde

茴香醛, 对甲氧基苯甲醛



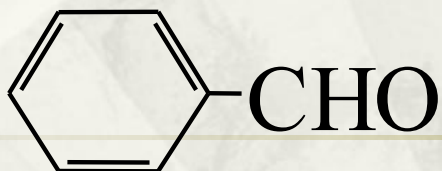
Furfuraldehyde

糠醛

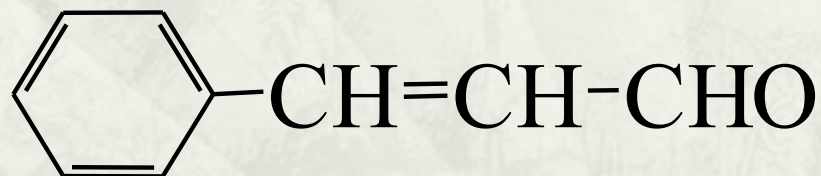


Furaldehyde

糠醛



Benzaldehyde



Cinnamaldehyde



Formaldehyde



Acetaldehyde

8.2 酮(ketones)

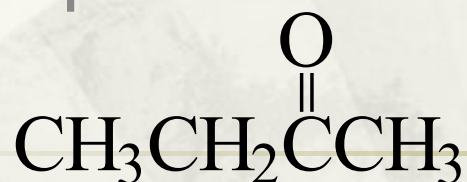
8.2.1 酮的命名

1) 酮的普通命名法是当 R_1COR_2 式中 R_1 和 R_2 相同时，在 R_1 的烃名前加“二”(di)，后面放一“酮”(ketone)字； R_1 和 R_2 不同时，将两者作为取代基按字母顺序排列，后面再加上“酮”(ketone)字，系统命名中，是将羰基中的碳原子作为链或者环系中的一员，将相应的烃名或环系后的词尾“e”改为“one”。编号时使羰基上的碳原子编号尽可能小。

(1). System name

Ketone = Alkan **-one**

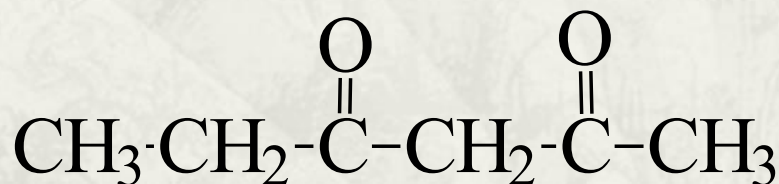
for example:



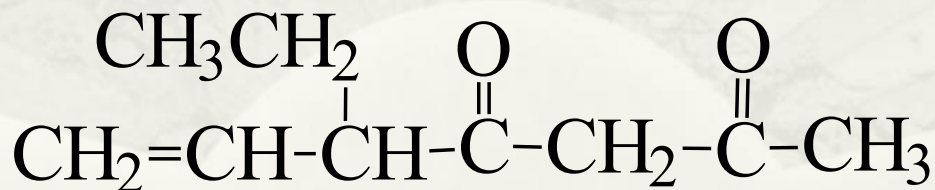
2-Butanone



4-Penten-2-one



2,4-Hexanedione



5-Ethyl-6-heptene-2,4-dione

根基命名法

Ketone = R + R' + ketone

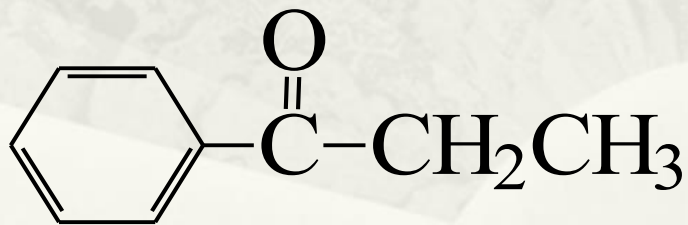
for example:



Diethyl ketone

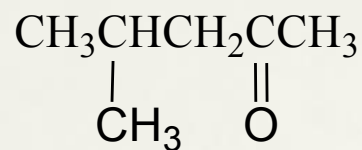


Ethyl methyl ketone

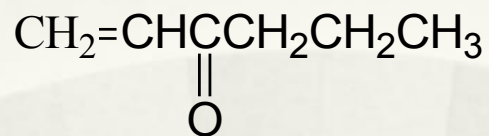


Phenyl ethyl ketone

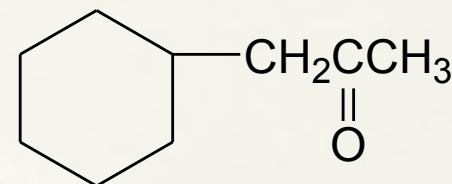
例如：



Isobutyl methyl ketone
4-Methyl-2-pentanone

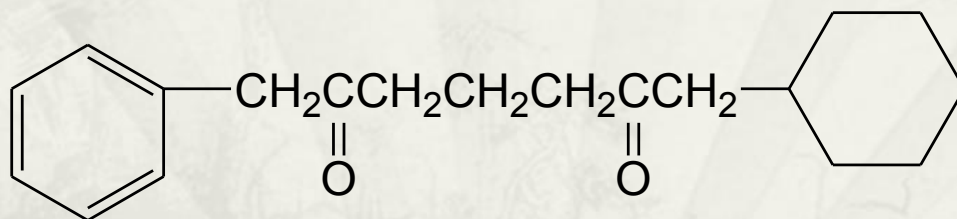


Propyl vinyl ketone
1-hexen-3-one



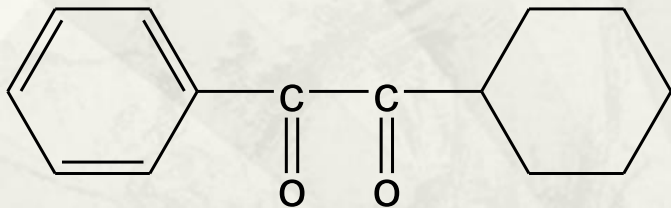
Cyclohexylacetone
Cyclohexyl-2-propanone

2) 脂肪酮与环系相连时，一般将环系作为脂肪酮的取代基按脂肪酮的系统命名法命名。
例如：

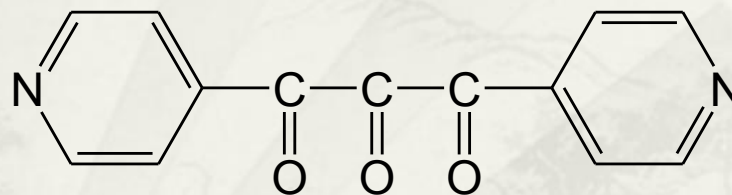


6-Cyclohexyl-1-phenyl-2,5-hexanedione

3) 含有两个或几个羰基直接相连的二酮、三酮等，当两端与两个环系相连时刻用根基命名法，英文名用diketone、triketone等。系统名则在相应碳原子数的烃名后加“dione”、“trione”等。例如：

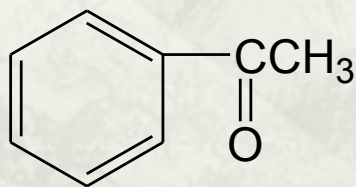


1-Cyclohexyl-2-phenylethanedione
Cyclohexyl phenyl diketone



Di-4-pyridylpropanetrione
Di-4-pyridyl triketone

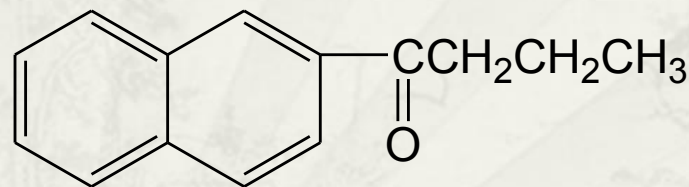
4) 环状化合物的单酰基衍生物还可将酰基作为词头放在环系名称前的办法命名，例如：



Acetophenone

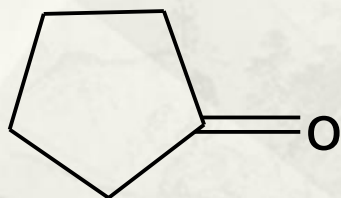
Acetobenzene

乙酰苯, 苯乙酮



2-Butyrylnaphthalene

5) 如羰基碳原子为环系的组成原子，根据羰基的多少可在环系名后加上“酮”、“二酮”或“三酮”等。例如：



Cyclopentanone



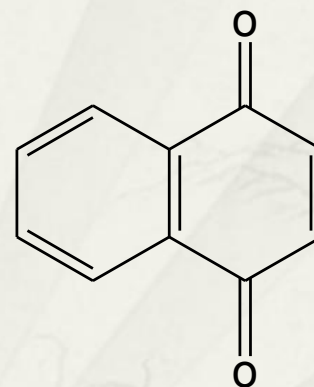
1,4-Cyclohexanedione

8.3 醌的命名

可将醌字(quinone)放在母体烃名前，也可按二酮的系统命名法命名。例如：

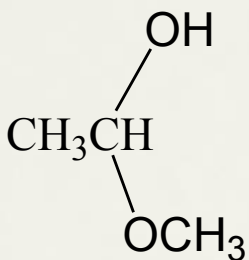


2,5-Cyclohexadien-1,4-dione
p-Benzoquinone

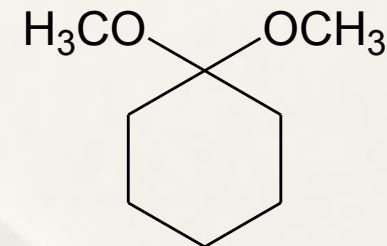


1,4-Naphthalenedione
1,4-Naphthoquinone

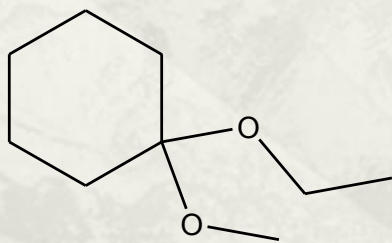
8.4 缩醛的命名



Acetaldehyde methyl acetal
1-Methoxyethanol



Cyclohexanone dimethyl acetal
1,1-Dimethoxycyclohexane

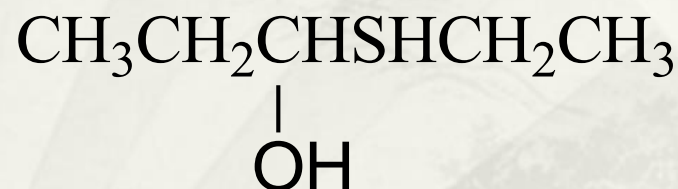


Cyclohexanone ethyl methyl acetal
1-Ethoxy-1-methoxycyclohexane

2) 缩醛中氧原子被硫原子取代后为硫缩醛，命名方法基本同缩醛。例如：

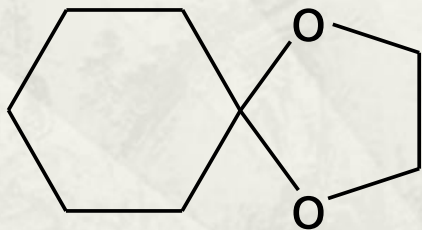


1,1-Bis(ethylthio)propane

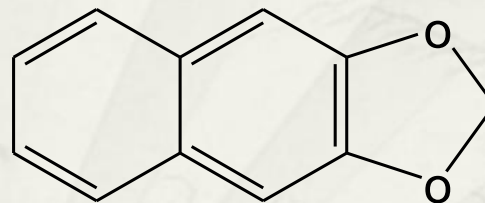


1-(Ethylthio)-1-propanol

3) 环状缩醛中，两个缩醛氧原子形成环的一部分，可按杂环化合物命名，或将 $\text{-OCH}_2\text{O-}$ 作为分子中的取代基而用词头“亚甲二氧基”（methylenedioxy）来命名。例如：



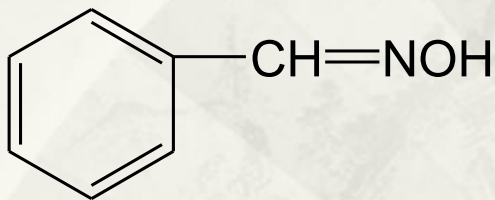
1,4-Dioxaspiro[4.5]decane



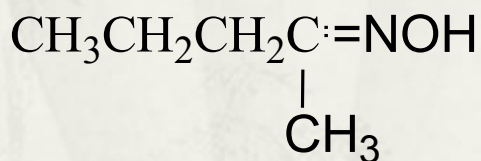
2,3-Methylenedioxy-naphthalene

8.5 肟的命名

1) $\text{RCH}=\text{NOH}$ 为醛肟, $\text{R}_1\text{R}_2\text{C}=\text{NOH}$ 为酮肟, 英文名称为 Oxime, 其命名时可用系统名称或在相应的醛或酮名称后加“肟”字。例如:



(Phenylmethylene)azanol
Benzaldehyde oxime



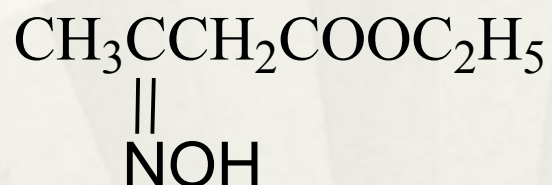
(1-Methylbutylidene)azanol
2-Pentanone oxime



Cyclohexanone oxime

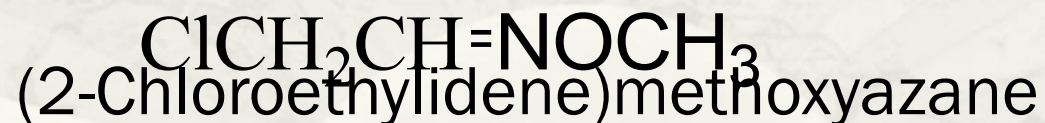
azanol : 羟胺

2) 肟分子中带有比羰基更优先的基团时，肟基可作为词头命名。例如：



Ethyl 3-(hydroxyimino)butyrate
hydroxyimino 肟基

3) =NOH中的氢原子被烷基取代后可按系统命名法命名，例如：

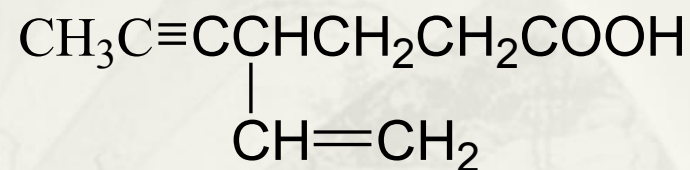
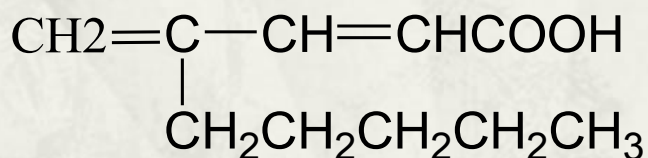


azane 氮烷；氮

9 羧酸及其衍生物

9.1 羧酸

1) 链状一元羧酸是在主链烃名后加一“酸”字，英文名称是将烃名后的“e”改为“-oic acid”，选择主链时，应为1)包括羧基的最长链，2)有不饱和键时选择含羧基和不饱和键的最长链。
例如：



Methanoic acid

4-Pentyl-2,4-pentadienoic acid

4-Vinyl-5-heptynoic acid

The carboxylic acid

(1). $-(C)OOH$ (The main chain involves carboxylic carbon)

Carboxylic acid = Alkan-oic acid

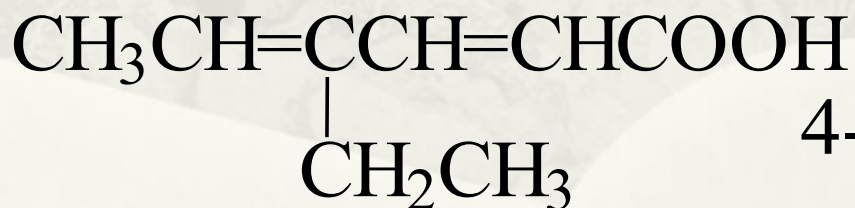
for example:



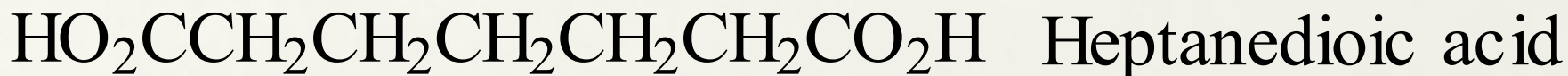
Hexanoic acid



3-Pentenoic acid



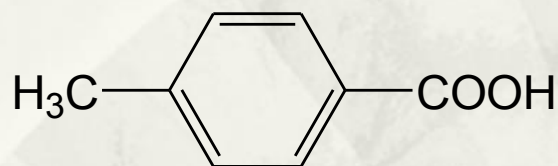
4-Ethyl-2,4-hexadienoic acid



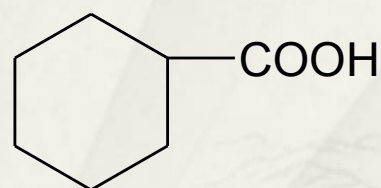
Heptanedioic acid

2) 连接命名法

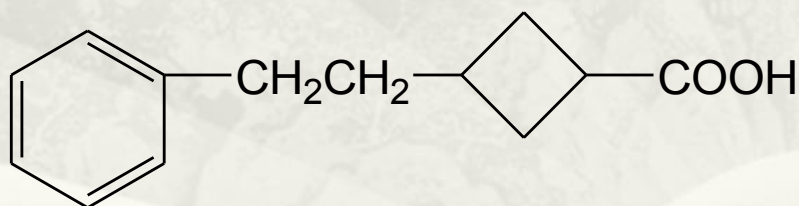
主要用于脂环、桥环、杂环和联环听的羧酸，编号时不含羧基碳原子，命名时在烃名后接“羧酸”（carboxylic acid）。
例如：



4-Methylbenzenecarboxylic acid



Cyclohexanecarboxylic acid

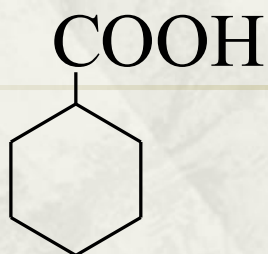


3-(2-Phenylethyl)-1-cyclobutanecarboxylic acid

(2). -COOH (The main chain without carboxylic carbon)

Carboxylic acid = Alkane-carboxylic acid

for example:



Cyclohexanecarboxylic acid



3-Cyclopentenecarboxylic acid



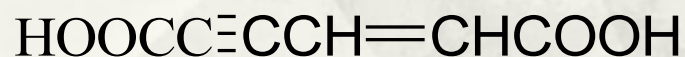
1,3,5-Pentatricarboxylic acid



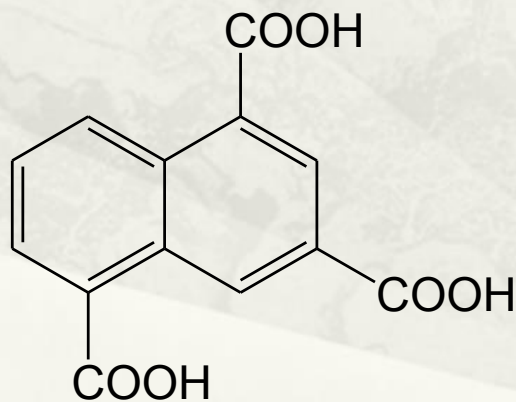
1,2,3-Butanetricarboxylic acid

9.2 多元羧酸

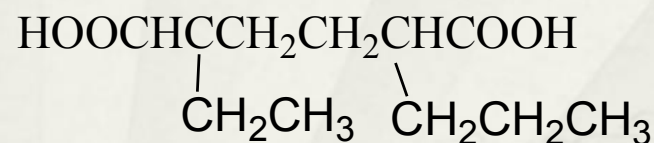
多元羧酸包括二元、三元等羧酸，命名原则同一元羧酸，须在“酸”前加数量词，英文名称在“oic”和“carboxylic”前加di、tri、tetra等。例如：



2-Hexen-4-ynedioic acid



1,3,5-Naphthalenetricarboxylic acid



4-Ethyl-2-propylhexanedioic acid

Acyl radicals

(1). $R(C)O\cdot$

Acyl radical = Alkan-oyl

for example:



Propanoyl



Hexanoyl

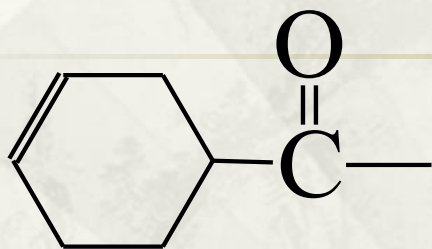


Heptanedioyl

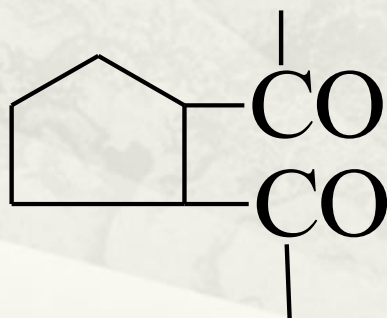
(2). RCO-

Acyl radical = Alkane-carbonyl

for example:



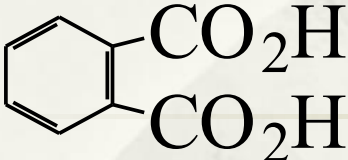
3-Cyclohexanecarbonyl



1,2-Cyclopentanededicarbonyl

(3). Some exception

	System	Trivial	RCO-
HCOOH	Methanoic	Formic	Formyl
CH_3COOH	Ethanoic	Acetic	Acetyl
$\text{HO}_2\text{CCO}_2\text{H}$	Ethanedioic	Oxalic	Oxalyl
$\text{HO}_2\text{CCH}_2\text{CO}_2\text{H}$	Propanedioic	Malonic	Malonyl
$\text{HO}_2\text{C}(\text{CH}_2)_4\text{CO}_2\text{H}$	Hexanedioic	Adipic	Adipoyl
$\text{HO}_2\text{C}(\text{CH}_2)_2\text{CO}_2\text{H}$	Butanedioic	Succinic	Succinyl

	System	Trivial	RCO-
	Benzenecarboxylic	Benzoic	Benzoyl
	1,2-Benzenedicarboxylic	Phthalic	Phthaloyl
$\begin{array}{c} \text{CHCO}_2\text{H} \\ \parallel \\ \text{CHCO}_2\text{H} \end{array}$	cis-butenedioic	Maleic	Maleoyl
$\begin{array}{c} \text{HO}_2\text{CH} \\ \parallel \\ \text{CHCO}_2\text{H} \end{array}$	trans-butenedioic	Fumaric	Fumaroyl

9.3 酰卤

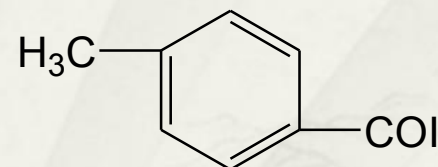
酰卤的命名是把酰基的名称后加上负卤离子名。英文名是把酸名的词尾“oic acid”和“ic acid”改为“oyl halide”或“yl halide”，使用连接法命名时把“carboxylic acid”改为“carbonyl halide”。例如：



Acetyl bromide



Acryloyl Chloride
(=acrylyl)丙烯酸酰



p-Anisoyl iodide
甲氧苯酰

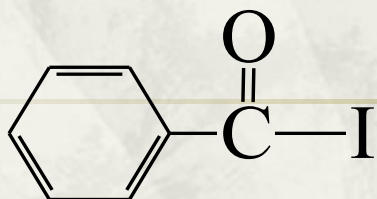


1,4-Benzenedicarbonyl dichloride

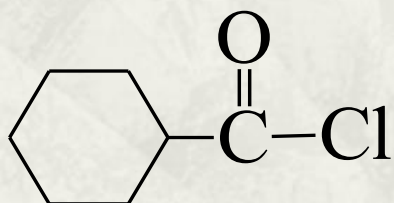
Acyl halides

Acyl halide = Alkane-carbonyl
or Alkan-oyl + halide

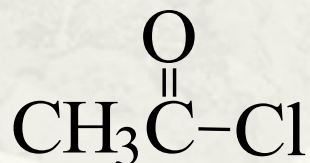
for example:



Benzoyl iodide



Cyclohexanecarbonyl chloride



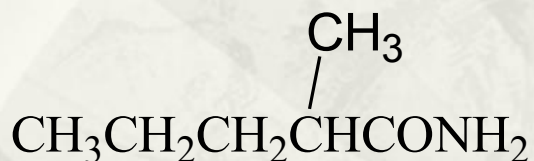
Acetyl chloride



Propanoyl chloride

9.4 酰胺

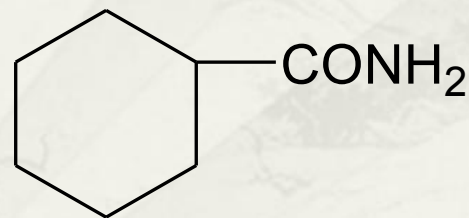
1) 酰胺的命名是把酸名的词尾“oic acid”和“ic acid”改为“amide”,若为“carboxylic acid”则改为“carboxamide”, 例如:



2-Methylpentanamide

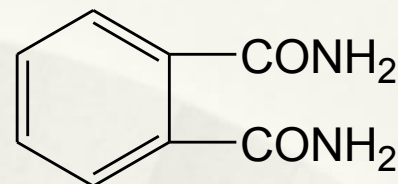
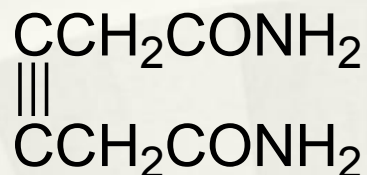
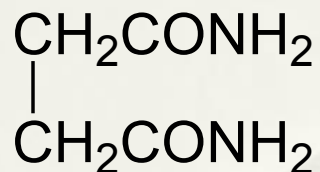


3-Butenamide



Cyclohexanecarboxamide

2) 多元酰胺的命名与多元羧酸相同。例如：



Butanediamide

3-Hexynediamide

1,2-Benzenedicarboxamide

3) 多元羧酸的部分羧基成为酰胺基后，因羧基比酰胺基的顺序优先，酰胺基作为取代基。例如：



Methyl 4-carbamoylcyclohexanecarboxylate

Carbamoyl 氨基甲酰

4) 酰胺分子中的氮原子上的氢原子被烃基取代后可作为N-取代的酰胺命名，当氮原子为环内原子时，则作为取代的杂环化合物命名。当酰基上所连烃基比氮原子上所连烃基简单时，可将酰基作为取代基命名。例如：



N-Phenylethanamide

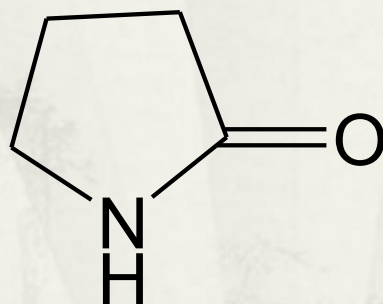


N,N-Dimethylformamide



2-(Diethylamino)-*N*-methylacetamide

5) 内酰胺可按杂环命名，也可在相同碳原子数目的烃名后加上“内酰胺”（Lactam）来命名。例如：



2-Pyrrolidone 或 4-Butanelactam

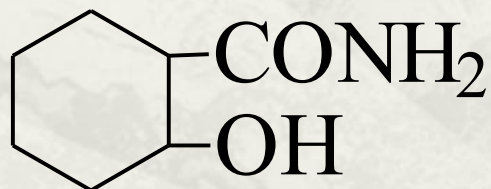
Amide

(1) Amide

A. -CONH_2 (The main chain without carbonyl carbon)

Amide = Alkane-carboxamide

for example:



2-Hydroxycyclohexanecarboxamide



1,3-Cyclopentadienecarboxamide

B. $-(\text{C})\text{ONH}_2$ (The main chain involves carbonyl carbon)

Amide = Alkan-**amide**

for example:



Butanamide



Ethanamide (Acetylamide)



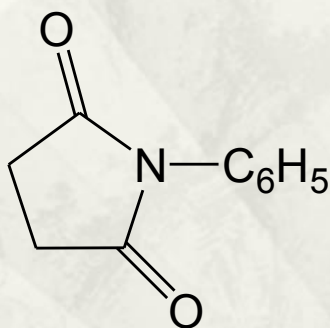
2-Methylpropanamide



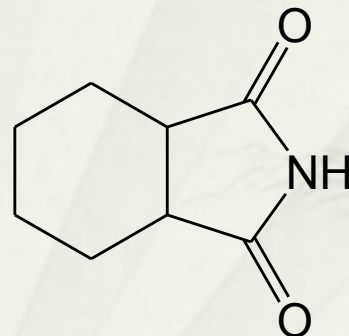
Hexanediamide

9.5 酰亚胺

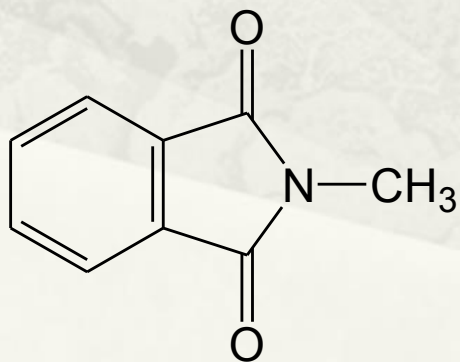
1) 酰亚胺命名时将相应羧酸词尾“二羧酸”改为“二酰亚胺”(dicarboxamide),或将普通名称词尾“ic acid”改为“imide”,也可按杂环命名。例如:



N-Phenylsuccinimide

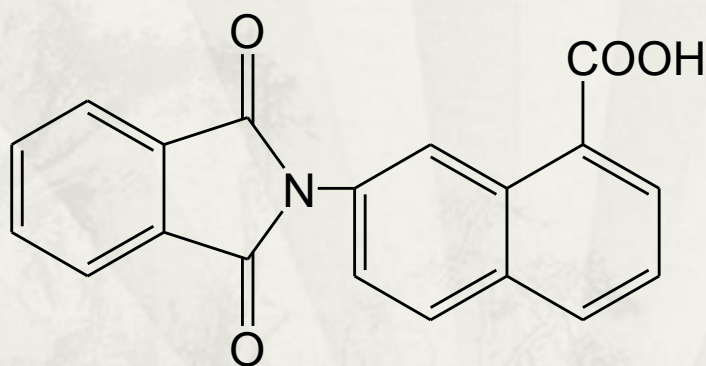


1,2-Cyclohexanededicarboximide



N-Methyl-1,2-benzenedicarboximide

2) 酰亚胺作为取代基时，酰亚胺基英文名称为“imido”，例如：

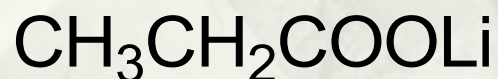


7-Phthalimido-1-naphthoic acid

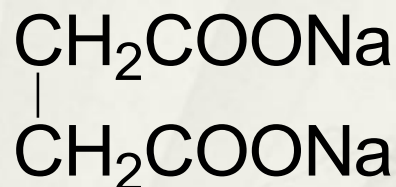
9.6 盐和酯

9.6.1 盐

- 1) 羧酸盐命名时在酸名后加上金属名即可，英文名称是把酸名的词尾“-ic acid”改为“-ate”，前面加上金属名。例如：

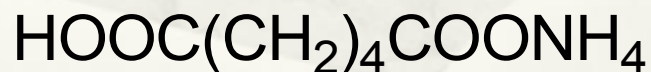


Lithium propanoate



Disodium butanedioate

- 2) 二元酸只有一个羧基成盐，命名时在酸和金属名之间加“氢”（hydrogen）字，多元酸的命名与酸相同，例如：



Amminium hydrogen hexanedioate

Carboxylic salts

Carboxylic salt = Metal + Alkan-oate

or Metal + Alkane-carboxylate

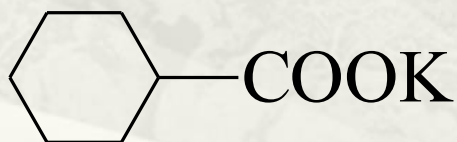
for example:



Potassium acetate



Sodium 2-butenate



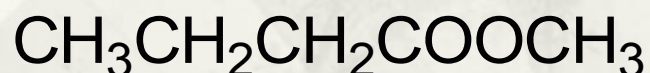
Potassium cyclohexanecarboxylate



Potassium sodium oxalate

9.6.2 酯

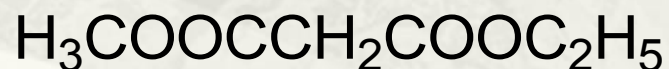
1) 酯的命名和盐类似，把盐中的金属名改为酯基中的烃名即可，多元酸酯基不同时，名称按字序排列。例如：



Methyl butyrate或Methyl butanoate

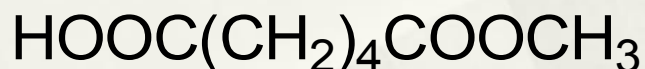


2-Chloroethyl chloroacetate

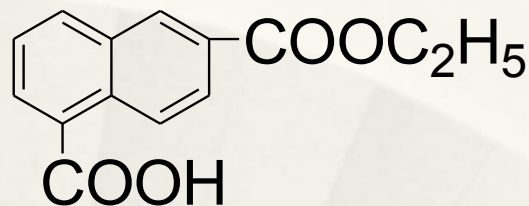


Ethyl methyl malonate (丙二酸)

2) 二元酸的单酯也可用加“酸性”(hydrogen)的办法命名。
例如：

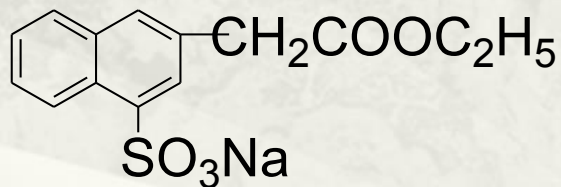


Methyl hydrogen adipate

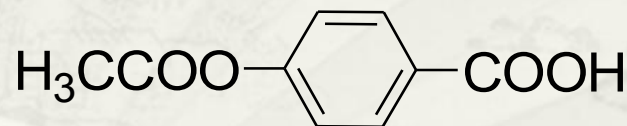


6-Ethyl hydrogen 1,6-naphthalenedicarboxylate

3) 化合物 R_1COOR_2 中，当 R_1 上连有优先基团、环烷或杂环时，可将 $-\text{COOR}_2$ 作为取代基命名，称为“烷氧羰基”。若 R_2 为母体， $\text{R}_1\text{COO}-$ 作为取代基命名，使用“酰氧基”。例如：

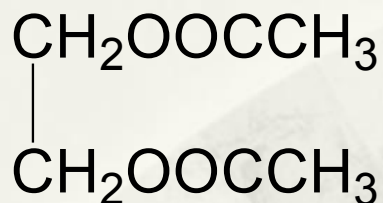


Sodium 3-(ethoxycarbonylmethyl)-
1-naphthalenesulfonate

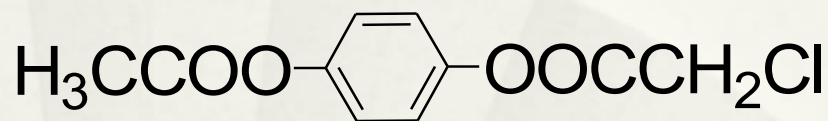


p-Acetoxybenzoic acid

4) 多元醇的酯命名时将醇名放在前面，酸名前加数词，不同的酸英文名称按字序排列。例如：

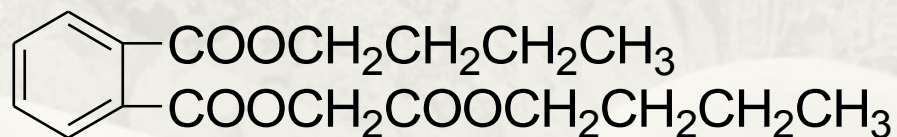


Ethylene diacetate



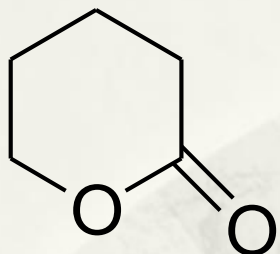
p-Phenylene acetate chloroacetate

5) 羟基酯和复杂的羧酸所形成的酯命名时将羟基酯作为取代基，例如：

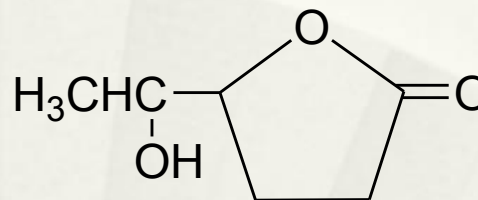


2-Butoxy-2-oxoethyl butyl benzenedicarboxylate

6) 内酯 内酯英文名是把词尾“-ic acid”改为“-olactone”,并标出定位号,也可用系统命名法,将烃链词尾“e”去掉,加上“-olide”,并标出与羧基关环的羟基的定位号。例如:

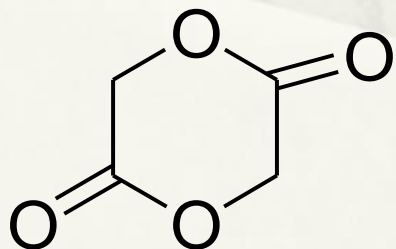


5-Pentanolide



4,5-Dihydroxyhexanoic-1,4-lactone

7) 交酯 英文命名时将酸的词尾“-ic acid”改为“-ide”,并根据酸分子的数目在前面相应加上数量词di、tri等。也可按杂环命名,例如:



Diglycollide或1,4-Dioxacyclohexane-2,5-dione

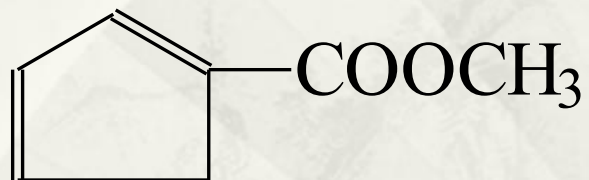
Esters

Ester = Alkyl + carboxylate (alkanoate)

for example:



Ethyl acetate



Methyl 1,3-cyclopentadienecarboxylate



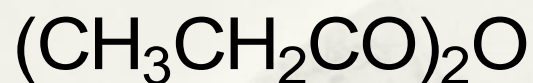
Diethyl malonate



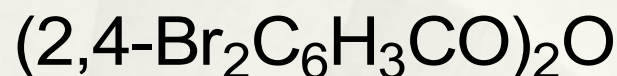
Isopropyl 2-propenoate (acrylate)

9.7 酸酐

- 1) 酸酐的英文名为“anhydride”,单酐命名时可在原酸名后加“酐”(anhydride)字,例如:



Propionic anhydride



Bis(2,4-dibromobenzoic)anhydride

- 2) 混酐英文命名时将酸名按字顺排列,后再加“酐”字。例如:

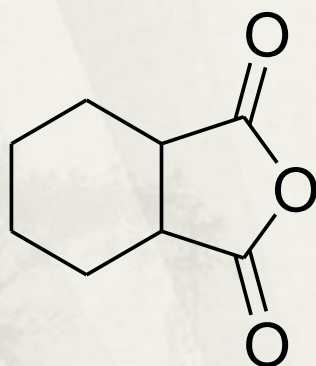


Benzoic propionic anhydride



Acetic chloroacetic anhydride

3) 由一分子二元酸或多元酸生成的酸酐，其具有杂环结构，但仍以酸酐命名。例如：



1,2-Cyclohexanedicarboxylic anhydride

Anhydrides

Anhydride = Carboxylic + anhydride

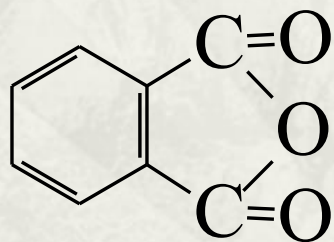
for example:



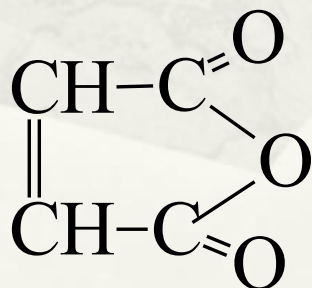
Propenoic anhydride



Acetic anhydride



Phthalic anhydride



Maleic anhydride

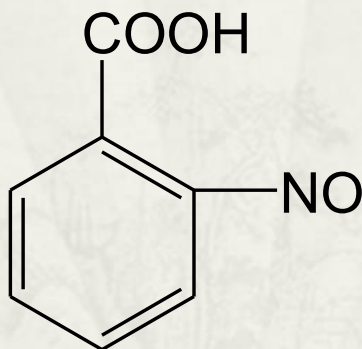
10 含氮化合物

10.1 硝基和亚硝基化合物

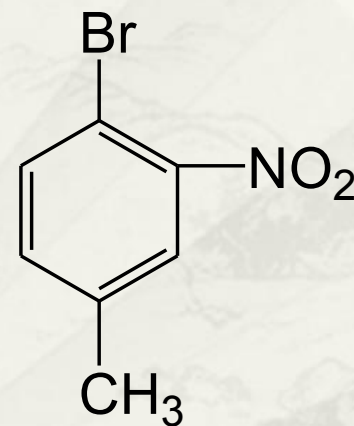
硝基（nitro）和亚硝基（nitroso）在命名中，只能作为词头用，只能采用取代命名法。例如：



Trichloronitromethane



2-Nitrosobenzoic acid

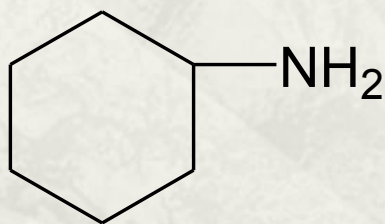


1-Bromo-4-methyl-2-nitrobenzene

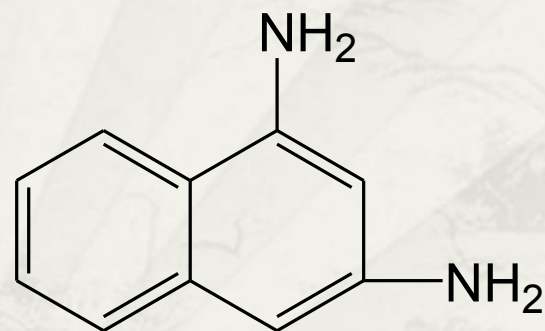
10.2 胺类和亚胺类化合物

1) 伯胺 (Primary amine)

一元伯胺通常是在烃基名后加上amine，二元或多元伯胺命名时将“二胺” (diamine)、 “三胺” (triamine) 等连在母体化合物或多价基的名称后。例如：



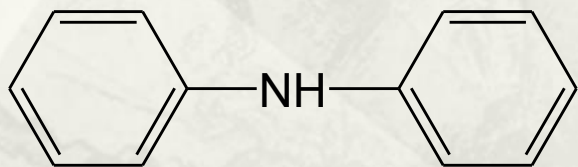
Cyclohexylamine



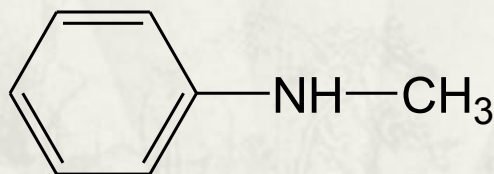
1,3-Naphthalenediamine

2) 仲胺和叔胺

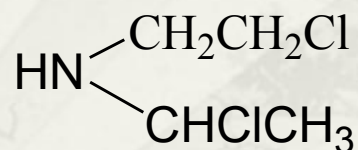
氮原子上连有相同的烃基时，可将“二”（di）、“三”（tri）加到烃基名称前，以胺作词尾来命名。如氮原子上的所连的烃基不同，则按优先顺序将烃基名称放在胺前。例如：



Diphenylamine



Methylphenylamine



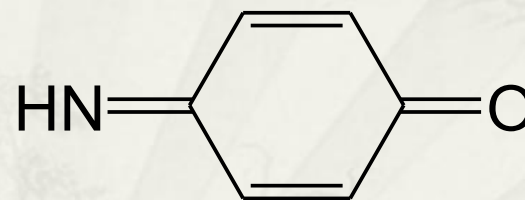
1-Chloro-N-(2-chloroethyl)ethylamine

3) 亚胺化合物

亚胺化合物命名是在烃名之后加上“亚胺”（imine）。如所带亚胺基不止一个时，则加“二亚胺”、“三亚胺”等，亚胺分子中有其他优先的官能团时，作为取代基而用词头“亚胺基”（imino）命名。



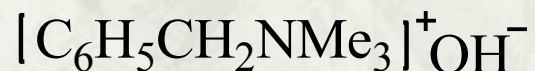
1,4-Cyclohexanediimine



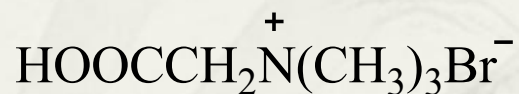
4-Imino-2,5-cyclohexadien-1-one

4) 季胺化合物

季胺化合物可按取代的铵盐或取代的氢氧化铵命名，英文名称把负离子名称放在最后，前面的取代基名后加 ammonium(铵)或 iminium(亚胺)，有些仍采用俗名。



Benzyltrimethylammonium hydroxide



Betaine hydrobromide

(2). Amine

A. Aliphatic amine

Amine = Alkyl + amine

for example:



Ethylenediamine



Propylamine



Triethylamine



N,N-dimethylbutylamine

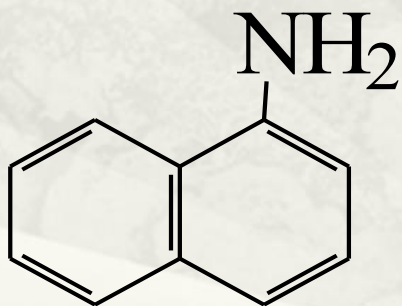
B.Aromatic amine

Amine = Aromatic compound + amine

for example:



Benzene amine



1-Naphthalene amine

10.3 脒、胍、脲

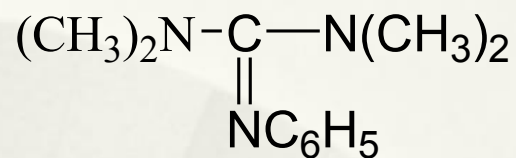
$\text{HC}(=\text{NH})\text{NH}_2$ 可用来表示脒 (amidine), $\text{HN}=\text{C}(\text{NH}_2)\text{NH}_2$ 表示胍 (guanidine), $\text{H}_2\text{N}-\text{C}(=\text{O})\text{NH}_2$ 表示脲 (urea)。

脒可用酸的命名法, 英文名称是把相应的酸的词尾 “-oic acid” 或 “-ic acid” 改为 “amidine”。把 carboxylic acid 改为 carboxamidine, 作为取代基时系统命名用 carbamimidoyl 做词头, 按惯例沿用 amidino。胍的衍生物按胍的取代产物命名, 编号时 $=\text{NH}$ 为 2, 两个 NH_2 分别编为 1 和 3, 作为取代基时为胍基 (guanidino)。脲的氮上的氢被取代后衍生物可按脲的衍生物命名, 可将碳原子编为 2, 两个氮原子分别编为 1 和 3, 作为取代基时为 “脲基” (ureido)。

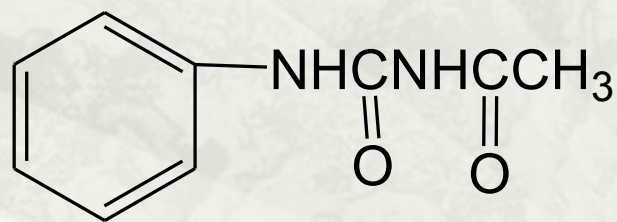
例如：



4-Ethylhexamidine



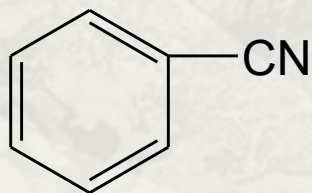
Tetramethyl-2-phenylguanidine



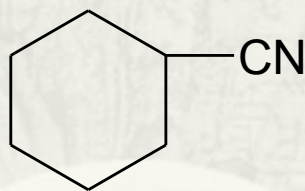
1-Acetyl-3-phenylurea

10.4 腈和异腈

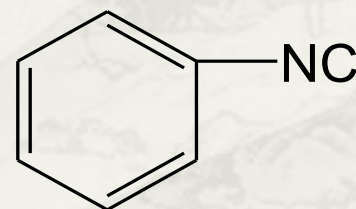
腈(nitrile)可有相应的酸的俗名来命名，英文名称将词尾-oic acid或-ic acid改为-onitrile。还可使用与酸相应的另一命名法，即把羧酸词尾“-carboxylic acid”改为“-carbonitrile”，-CN作取代基时为腈基（cyano）。异腈命名时在烃基名后加“isocyanide”，作为取代基时用词头“异腈基”（isocyano）
例如：



Benzonitrile



Cyclohexanecarbonitrile



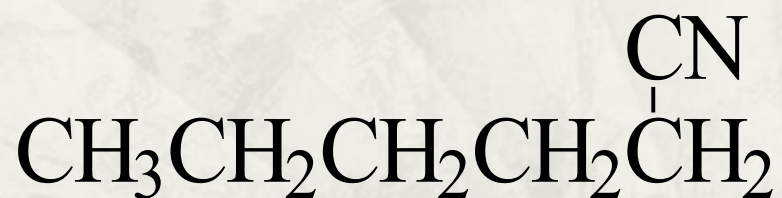
Phenyl isocyanide

Nitriles

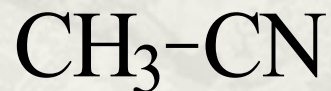
(1) R-(C)N (The main chain involves cyanocarbon)

Nitrile = Alkane + nitrile

for example:



Hexanenitrile



Ethanenitrile

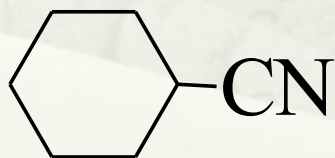
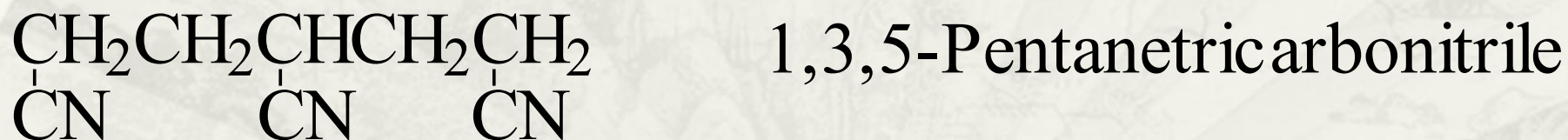


3-Methylbutanenitrile

(2) R-CN (The main chain without cyanide carbon)

Nitrile = Alkane-carbonitrile

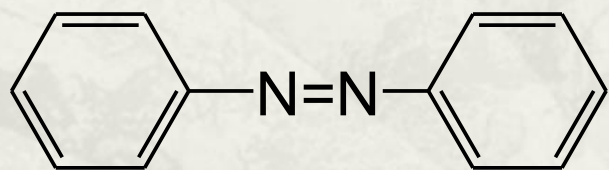
for example:



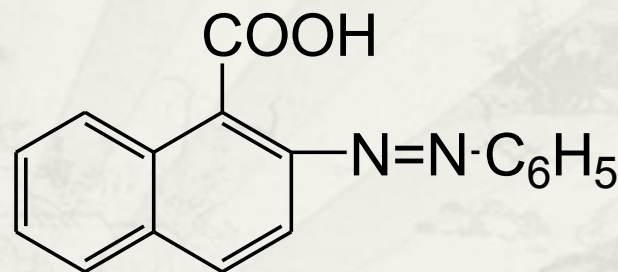
Cyclohexanecarbonitrile

10.5 偶氮、重氮和叠氮化合物

1) 偶氮基-N=N-两端连有烃基的化合物称为偶氮化合物 (azo-compound)。两个烃基相同时，把“偶氮”作为词头放在烃名前，将与-N=N-相连的原子编号放在最前面，可用带撇的数字区别。两端烃基不同时，自较复杂的烃基开始依次向另一端命名。

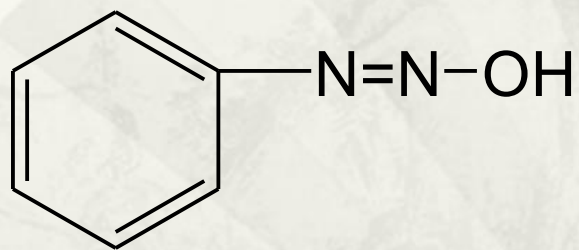


Azobenzene

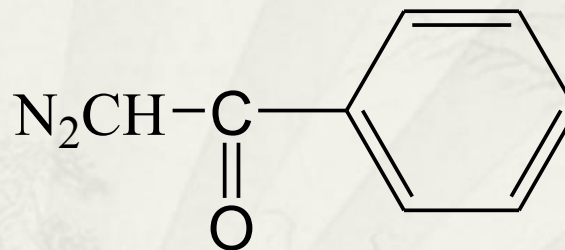


1-Carboxynaphthalene-2-azobenzene

2) 重氮基-N=N-只有一端与芳烃基相连或=N≡N与一二价烃基相连而成的化合物R-N=N_x (X=OH、CN等) 和>C=N₂, 前者英文名将“diazo”和X的基名作为词尾放在烃名后。后者是将“重氮”作为词头放在母体名称之前。

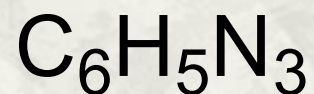


Benzenediazohydroxide

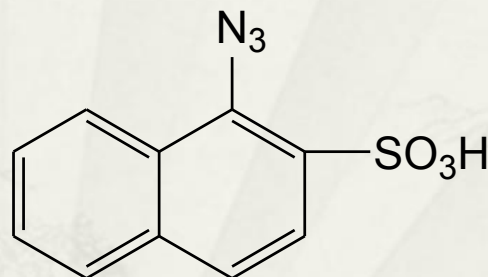


Ethyl diazoacetate

3) 含有叠氮基 ($-N_3$) 的化合物可在烃名后加上“叠氮” (azide) 来命名, 或按取代命名法将“叠氮基” (azido) 作为词头来命名。例如:



Phenylazide or
Azidobenzene



1-Azido-2-
naphthalenesulfonic acid

系统命名中主要官能团的优先顺序

1. 鎓盐 (onium and similar cations)
2. 酸 (acids) : 按COOH、(C)OOH次序, 然后是它们的S和Se的衍生物, 再次为磺酸、亚磺酸等。
3. 酸的衍生物 (Derivatives of acids) :按酸酐、酯、酰卤、酰胺、酰肼、酰亚胺、脒等的次序。
4. 腈, 异腈 (nitriles, isocyanides)
5. 醛 (aldehydes), 然后是他们的S和Se的衍生物, 再次是它们的衍生物。
6. 酮 (ketones), 其类似物和衍生物次序与醛同。
7. 醇 (alkohols), 次为酚 (phenols) 然后是它们的S、Se和Te的类似物。
8. 氢过氧化物 (hydroperoxides)
9. 胺 (amines), 然后是亚胺 (imines) 和肼 (hydrazines) 等
10. 醚 (ethers), 其次为它们的S和Se的类似物
11. 过氧化物 (peroxides)

一些作词头的常见官能团

Characteristic group	prefix	Characteristic group	prefix
-Br	bromo	-IO ₂	iodyl
-Cl	chloro	-I(OH) ₂	dihydroxyiodo
-ClO	chlorosyl	-N ₂	diazo
-ClO ₂	chloryl	-N ₃	azido
-ClO ₃	perchloryl	-NO	nitroso
-F	fluoro	-NO ₂	nitro
-I	iodo	-OR	R-oxy
-IO	iodosyl	-SR	R-thio

主要官能团作词头和词尾时的应用名称

Class	Formula	Prefix	Suffix
carboxylic acid	-COOH	carboxy	-carboxylic acid
acid	-(C)OOH	-	-oic acid
sulfonic acid	-SO ₃ H	sulfo	-sulfonic acid
salts	-COOM -(C)OOM	--- ----	metal...carboxylate metal...oate
esters	-COOR -(C)OOR	R-oxycarbonyl ----	R-carboxylate R-carboxylate
acid halides	-CO-halogen -(C)O-halide	Haloformyl ----	-carbonyl halide -oyl halide
amides	-CO-NH ₂ -(C)O-NH ₂	Carbamoyl -----	-carboxamide -amide
amidines	-C(=NH)-NH ₂ -(C)(=NH)-NH ₂	Amidino 172 ---	-carboxamidine -amidine

Class	Formula	Prefix	Suffix
nitriles	-C≡N -(C)≡N	Cyano ----	-carbonitrile -nitrile
aldehydes	-CHO -(C)HO	formyl oxo	-carbaldehyde -al
ketones	=(C)=O	oxo	-one
Alcohols and phenols	-OH	hydroxy	-ol
thiols	-SH	mercapto	-thiol
amines	-NH ₂	amino	-amine
imines	-NH	imino	-imine
ethers	-OR	R-oxy	--
sulfides	-SR	R-thio	--

**Halohydrocarbon = Alkyl + halide
or prefix-alkane**

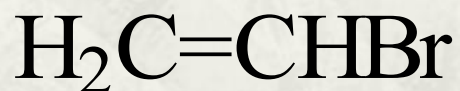
For example:



Carbon tetrachloride



Methylene chloride

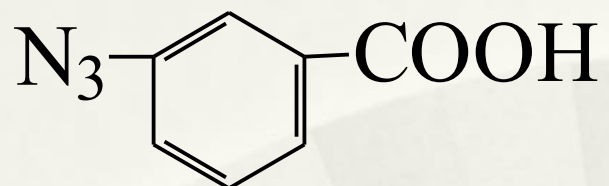


Vinyl bromide

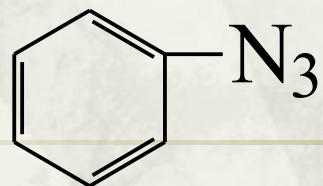


1,2-dichloropropane

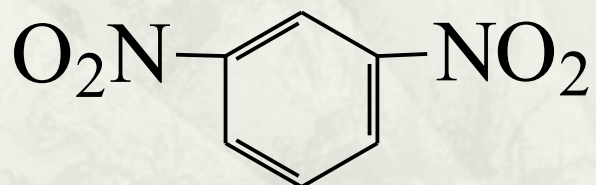




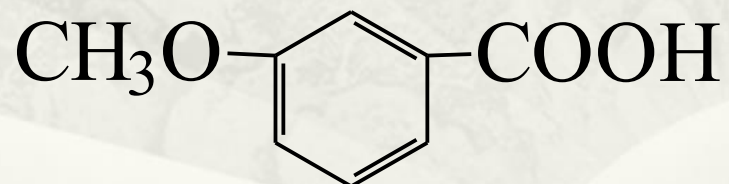
3-Azidobenzoic acid



Phenyl azide



1,3-Dinitrobenzene



3-Methoxybenzoic acid

The characteristic group in the order for citation

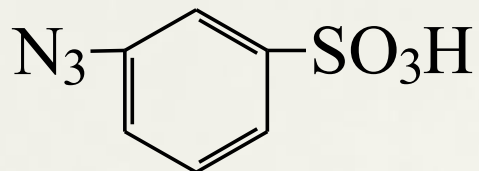
(1) $\text{COOH} > \text{general acid} > -\text{SO}_3\text{H}$

(2) $\text{RCO-O-OCR}' > -\text{COR} > -\text{COX} > -\text{CONH}_2$

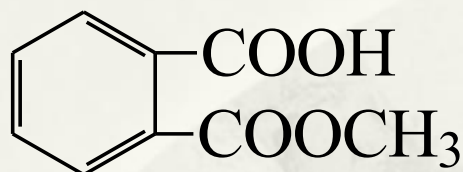
(3) $-\text{CN} > -\text{CHO} > \text{C=O}$

(4) $-\text{OH} > \text{NH}_2 > \text{NH} > -\text{NH-NH}_2- > \text{RO-}, \text{N}_2, \text{X}$

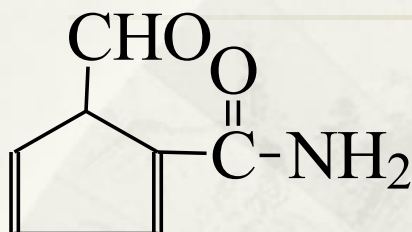
For example:



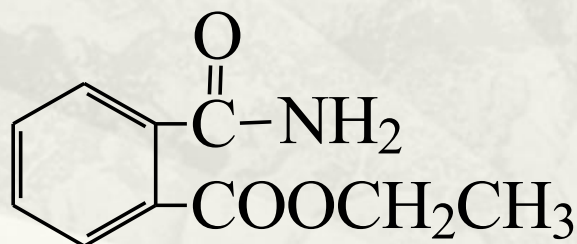
3-Diazobenzenesulfonic acid



2-Methoxycarbonylbenzoic acid



5-Formyl-1,3-cyclopentadienecarboxamide



Ethyl 2-Carbamoylbenzoate



7-Hydroxy-2-heptanone

Exercise

